ACL_TEAM_61 / MILESTONE 1

Formula 1 Podium Prediction Companion

Analytical report

From relational race data to pre-race podium probabilities

Final source: **Kaggle Version 13**
Saved execution: **successful, 127.6 seconds**
Report prepared: **16 October 2026**

Release evidence	Verified result
Notebook structure	151 cells: 125 Python + 26 Markdown
Execution completeness	Python execution counts 1-125, in order; no error cells
Figures	42 original output images retrieved and included
Selected model	Shallow neural network; validation AP 0.7496; threshold 0.28
Scope	Report preparation only; no training, split, hyperparameter or sharing changes

The milestone description defines the grading and submission requirements [1]. The beginner guide supplies Formula 1 context [2]. All dataset counts, model metrics, statistical estimates and figures in this report come from the **final successfully executed Version 13** [3]. Earlier notebook versions are not numerical sources.

Saved viewer:
https://www.kaggle.com/code/amrelkotit/acl-team-61?scriptVersionId=359134370

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Reading the report. Sections 1-16 provide the analysis and interpretation. Appendix A reproduces all 42 original notebook figures with captions, source cell numbers and concise interpretations. Appendix B maps the requirements to evidence and records the remaining methodological limitations. Tables use the values printed by Version 13; small apparent differences may reflect its displayed rounding.

Source locator convention. References such as [3, cell 128] identify the 1-based notebook cell, counting both code and Markdown cells. They are different from the Python execution counter. Appendix figures retain notebook output order.

1. Executive summary and prediction objective

The task is binary classification for one driver entering one race: estimate whether the driver will finish on the podium. The target is **positionOrder <= 3**, with one modeling row per **(raceId, driverId)**. The prediction cutoff is after qualifying and before the main race starts [1, pp. 1-3]. Current-race results, points, status, laps, pit stops, finishing times and post-race standings are excluded from predictors.

The pipeline validates the 14-table source, resolves historical shared-drive duplicates, constructs past-only form and reconstructed season-points features, and compares Logistic Regression (LR), Random Forest (RF) and a shallow neural network (NN) on the same 16 predictors and chronological rows. The primary metric is average precision (AP), labeled PR-AUC in the notebook; ROC-AUC and threshold-dependent F1, precision and recall support interpretation. Accuracy is supplementary.

The NN was selected using **validation AP 0.7496**, with a validation-F1-selected probability threshold of **0.28**. On 2022-2024 test data it achieved **ROC-AUC 0.9299, AP 0.6847 and F1 0.6381**. LR achieved higher test AP (0.7063), but test performance does not replace the validation-selected NN. [3, cell 126] [3, cell 128]

The engineering answers show circuit- and era-dependent front-row conversion, no established positive home-country effect after grid-band adjustment, and lower broad-classification recorded mechanical-retirement rates in the later era. The home-effect interval includes zero, and the leading later-era constructor changes under conservative status coding. These are descriptive associations, not causal estimates.

All assertions passed and both complete-record and partially missing-record inference executed. Important limits remain: historical entry/start ambiguity, changing qualifying coverage, retrospective source corrections, reconstructed rather than official standings, validation reuse, and prior inspection of the test period. The notebook is a reproducible executed study, but cannot retrospectively be characterized as a pristine untouched-test experiment.

2. Dataset and relational structure

The attached Kaggle/Ergast snapshot spans 1950-2024 and contains 1,125 races, 861 drivers, 212 constructors and 26,759 raw result rows. These are source-level descriptive counts, not training counts. The raw row counts and candidate primary keys are shown below. [3, cell 7] [3, cell 9]

Table	Raw rows	Primary key	Role
circuits	77	circuitId	Circuit names, geography and home-country matching
constructors	212	constructorId	Constructor identity and names
constructor_results	12625	constructorResultsId	Constructor-level recorded race results; audited only
constructor_standings	13391	constructorStandingsId	Official post-race standings; audited, not used directly as predictors
drivers	861	driverId	Driver identity and nationality
driver_standings	34863	driverStandingsId	Official post-race standings; audited, not used directly as predictors
races	1125	raceId	Race calendar, season, round, date and circuit joins
results	26759	resultId	Driver-race entries, outcome labels and earlier-race histories
qualifying	10494	qualifyId	Qualifying positions and session timing
lap_times	589081	raceId, driverId, lap	Race lap records; audited, excluded from current-race features

Table	Raw rows	Primary key	Role
pit_stops	11371	raceId, driverId, stop	Race pit-stop records; audited, excluded from current-race features
sprint_results	360	resultId	Sprint results; audited, not selected as features
seasons	75	year	Season-year parent relation
status	139	statusId	Retirement/status classification for retrospective Question 3

Key join paths are results - races - circuits for Question 1; results - drivers - races - circuits for Question 2; and results - races - constructors - status for Question 3. Qualifying joins on (raceId, driverId). Many-to-one and one-to-one merge validation prevents silent multiplication of records. The correct season reference is **races.year -> seasons.year**.

3. Initial EDA and its implications

The raw target has **3,397 podium rows (12.69%)** and **23,362 non-podium rows (87.31%)**. A majority-only rule can therefore look accurate while detecting no podiums. AP is prioritized because it assesses precision-recall ranking under class imbalance. Seasonal race counts, missingness, historical duplicate patterns and era distributions are plotted in Figures A1-A11. [3, cell 18]

Historical grid association is strong: podium rates are **63.47% at P1, 55.64% at P2, 46.28% at P3 and 5.84% at P10**. P1-P2 jointly convert at **59.58%**, compared with **8.97%** for other positive grid positions. The initial grid EDA uses 25,121 raw rows and excludes 1,638 non-positive or missing grid observations. These raw historical rates motivate grid as a predictor but do not isolate a causal grid effect. [3, cell 21] [3, cell 23]

Period	Mean annual qualifying-record coverage (%)
1950-1995	4.093224
1996-2005	50.542256
2006-2013	99.786184
2014-2021	99.520586
2022-2024	100.000000

These are unweighted means of annual driver-race qualifying-record coverage, not the completeness of every Q1/Q2/Q3 field. The fixed primary training start at 2006 reduces historical format and coverage drift. It is not an empirically demonstrated optimum against every possible earlier training window. [3, cell 28]

Driver and constructor podium-rate EDA supports competitiveness/history hypotheses (Figures A6-A7). For example, the displayed raw rates include Juan Fangio 68.63%, Mercedes 45.71% and Brawn 44.12%. Historical driver plots combine distinct race denominators with raw podium-row numerators and may include shared-drive duplicates. They are descriptive motivation, not direct measures of isolated skill. [3, cell 30] [3, cell 32]

4. Cleaning, typing, integrity and preprocessing impact

4.1 Immutable sources and semantic missingness

Raw CSVs remain unchanged. Cleaning operates on deep copies. The literal \N sentinel is recognized on import; race and session dates are parsed into datetimes; Q1/Q2/Q3 durations are converted to seconds; and selected numerical fields are coerced safely. Missing later qualifying sessions can be structural rather than erroneous. Nullable dates, lap times and finishing positions are measurements, not permission for missing required identifier references.

4.2 Primary keys, foreign keys and driver-race grain

All **14 table primary keys** are unique and non-null, including lap_times (raceId, driverId, lap), pit_stops (raceId, driverId, stop), and sprint_results.resultId. All **23 declared required FK relationships** passed with zero missing identifiers and zero orphan references. Required FK failures raise assertions. No nullable FK exception is assumed for these declared relationships. [3, cell 9] [3, cell 57] [3, cell 58]

There are **176 rows in 85 repeated driver-race groups**, representing **91 excess rows**. The deterministic shared-drive rule prefers a positive grid position, then the lowest resultId, retaining one row per driver-race. The cleaned result table contains **26,668 rows**. This preserves the required prediction grain while simplifying historical multi-car/shared-drive detail. [3, cell 48] [3, cell 50]

Measure	Before cleaning	After cleaning
Total Records	26759.00	26668.00
Duplicate Driver-Race Records	91.00	0.00
Podium Rate (%)	12.69	12.67

The expected cleaned count is derived from the source snapshot's distinct (raceId, driverId) pairs, rather than a fixed version-specific total. SHA-256 CSV fingerprints identify the snapshot; the final audit reconfirms unchanged raw files. Cleaned exports were checked for matching row counts and columns. [3, cell 51] [3, cell 61] [3, cell 149]

4.3 Timing anomaly and transformation impact

One 1995 Japanese Grand Prix Q1 value, **16:42.640 = 1,002.640 seconds**, is extreme relative to its same-race median **101.4735 seconds**. The source value is retained for audit. Only the modeling copy is flagged and set missing; no replacement lap is invented. The original and trimmed distributions are shown in Figures A13-A14. The cause remains independently unverified. [3, cell 62] [3, cell 63]

Median imputation and scaling are fitted on training data only. Figures A12-A14 and A28 show the effects of deduplication, semantic conversion and numerical transformation. Scaling changes location and range, not the underlying sporting interpretation or necessarily skewness. Missingness indicators distinguish unavailable qualifying measurements from real values.

5. Question 1: front-row conversion by circuit and era

Answer. Among circuits with at least 10 front-row observations, Circuit of the Americas leads at **20/24 = 83.33%**, followed by Sochi at **13/16 = 81.25%** and Magny-Cours at **29/36 = 80.56%**. Front row means actual recorded grid P1/P2; the numerator is a final positionOrder of 1-3. The analysis joins results to races and circuits at the deduplicated driver-race grain. [3, cell 66] [3, cell 67]

Circuit	Front-row observations	Podiums	Conversion (%)
Circuit of the Americas	24	20	83.333333
Sochi Autodrom	16	13	81.250000
Circuit de Nevers Magny-Cours	36	29	80.555556
Yas Marina Circuit	32	25	78.125000
Suzuka Circuit	68	51	75.000000

Era comparisons require at least 10 front-row observations in **both** pre-2014 and 2014-2024 periods; 16 circuits qualify. Suzuka changes from 35/50 (70.00%) to 16/18 (88.89%); Yas Marina from 6/10 (60.00%) to 19/22 (86.36%); and Monaco from 59/120 (49.17%) to 17/20 (85.00%). These comparisons show descriptive ranking shifts, not a causal regulation effect. Figures A15-A18 support the answer. [3, cell 69] [3, cell 70]

Qualifying position and starting grid are not interchangeable. Among **10,407** positive-position matched records, **2,422 (23.27%)** differ. On the comparable subset, actual grid-front-row entries convert at **70.18% (692/986)** versus **68.97% (680/986)** for qualifying-front-row entries. The two groups have equal counts here but need not contain identical drivers. A difference is not automatically a penalty; sprint procedures and source conventions also matter. [3, cell 72] [3, cell 73]

Sensitivity checks retain 2,250 primary front-row entries, 2,227 after excluding Indianapolis, 2,248 after excluding the unusual 2005 US GP, and 2,225 after excluding both. Exclusion can change rates and sample eligibility. Unequal circuit histories, source conventions, race dependence and small denominators limit ranking certainty; no statistical ranking confidence is claimed.

6. Question 2: home-country podium effect after grid control

Answer. The grid-band-standardized home-minus-away difference is **-0.528623 percentage points**. The approximate race-cluster bootstrap 95% interval is **[-1.999187, 0.754764] pp**, which includes zero. The data do not establish a positive home-country podium advantage after the implemented adjustment, nor a statistically established home disadvantage. [3, cell 81] [3, cell 82]

The multi-hop join links results, driver nationality, race and circuit country. Nationalities are harmonized; six unmapped driver records remain unassigned rather than guessed. Restriction to positive grid positions leaves **25,016 observations**: 2,322 home and 22,694 away. Raw home and away podium rates are **11.20%** and **13.74%**; these are composition-sensitive. [3, cell 77] [3, cell 78] [3, cell 79]

Home and away rates are compared within P1-P2, P3-P5, P6-P10, P11-P15 and P16+ bands and standardized using the same pooled grid-band weights. Adjusted home and away rates are **13.02%** and **13.55%** at the printed precision. Figures A19-A21 show the raw, within-band and uncertainty comparisons.

Sensitivity scenario	Rows	Home entries	Adjusted difference (pp)
Original sample	25016	2322	-0.529
Exclude possible non-starters	24912	2310	-0.521
Exclude Indianapolis 500	24635	1942	-0.465
Exclude both	24531	1930	-0.451

The interval uses 300 valid complete-race bootstrap draws across 1,125 race clusters, with fixed pooled grid-band weights. This preserves within-race dependence but does not fully model repeated drivers across races. Grid bands leave within-band residual confounding; driver/team quality, era shifts and selection into home races remain uncontrolled. Post-race status is used only in the non-starter sensitivity analysis, not as a predictor.

7. Question 3: constructor mechanical-retirement rates

Answer under the broad documented classification. The overall eligible-record rate is **303/3,258 = 9.30%** in 2014-2021 and **75/1,354 = 5.54%** in 2022-2024. The earlier leaders are Caterham, Lotus F1 and Toro Rosso; the later leaders are Alfa Romeo, Alpine and AlphaTauri. [3, cell 94]

Era	Constructor	Eligible starts (proxy)	Mechanical	Rate (%)
2014-2021	Caterham	34	8	23.53
2014-2021	Lotus F1	75	12	16.00
2014-2021	Toro Rosso	242	38	15.70
2022-2024	Alfa Romeo	88	10	11.36

Era	Constructor	Eligible starts (proxy)	Mechanical	Rate (%)
2022-2024	Alpine F1 Team	136	15	11.03
2022-2024	AlphaTauri	88	6	6.82

The join is results - races - constructors - status. The 2014-2024 sample has 4,626 records: 378 broadly classified mechanical retirements, 14 known non-starters and 4,234 other outcomes. Known non-starters are removed from the denominator, and main rankings require at least 20 eligible records per constructor per era. Finished and lapped statuses are not mechanical retirements; explicit accident, driver-error and regulatory outcomes are outside the mechanical numerator. Figures A22-A25 show the original rates, final rates and sensitivity results. [3, cell 86] [3, cell 93] [3, cell 94]

The broad mapping includes component/system labels such as Engine, Gearbox, Brakes, Hydraulics, Electrical, Power Unit, ERS, leaks and pressure/pump failures. Generic Mechanical, Technical, Vibrations and Wheel labels are treated as ambiguous in the sensitivity analysis, alongside several uncertain-cause or start/administrative labels. Power loss (13 records) remains outside the broad numerator. A component label can hide collision-induced damage; the actual start and root cause cannot always be inferred from status alone.

2022-2024 classification	Leader	Count / denominator	Rate (%)
Original broad mapping	Alfa Romeo	10 / 88	11.36
Ambiguous causes removed from numerator	Alpine F1 Team	14 / 136	10.29
Ambiguous records removed entirely	Alpine F1 Team	14 / 135	10.37

Under the numerator-only conservative rule, Alfa Romeo is second at 9/88 = 10.23%; under full ambiguous-record exclusion it is second at 9/87 = 10.34%. The rank reversal shows why status coding must accompany the answer. Caterham remains the earlier-era leader. Constructor rebranding is not retrospectively merged; unequal era windows and eligible-record proxies constrain interpretation. Lower later-era rates do not establish a causal effect of regulation changes. [3, cell 95]

8. Feature selection, derivation and leakage prevention

All 16 selected numeric features are retained. Their justification combines sporting hypotheses, the original EDA, training-only association/coverage plots and validation group ablation. No test metric determines a feature change. Figures A26 and A29-A30 show feature missingness, training associations and temporal coverage diagnostics. Outcome labels and retrospective status remain outside the feature matrix. [3, cell 97] [3, cell 103] [3, cell 104] [3, cell 112]

Feature	Source	Derivation and cutoff availability	Evidence / hypothesis
grid	results	Recorded grid, including raw zero convention; published final grid assumed known.	Grid EDA; grid ablation
grid_missing	results	1 for null or non-positive grid; known from pre-race grid.	Special grid convention, not a rank ahead of pole
qualifying_position	qualifying	Joined driver qualifying position; completed qualifying.	Qualifying/grid comparison; group ablation
q1_seconds	qualifying	Parsed Q1 duration; flagged anomaly missing in modeling copy.	Pace proxy; training distribution
qualifying_available	qualifying	Indicator that qualifying position is present.	Historical coverage drift
q1_missing	qualifying	Indicator that modeled Q1 duration is missing.	Separates missing timing from real duration

Feature	Source	Derivation and cutoff availability	Evidence / hypothesis
driver_prior_starts	results + races	Count of strictly earlier unique driver-race entries.	Exposure hypothesis; driver-form ablation
driver_career_podium_rate	results + races	Prior podium count / prior entries, excluding current race.	Persistent competitiveness hypothesis
driver_last5_podium_rate	results + races	Mean podium outcome over five preceding recorded races.	Recent-form hypothesis
constructor_prior_entries	results + races	Prior constructor-race entry totals.	Team exposure hypothesis
constructor_career_podium_rate	results + races	Prior team podiums / prior entries; aggregate before exclusion.	Historical team strength
constructor_last5_podium_rate	results + races	Prior five constructor races: summed podiums / summed entries.	Recent team-strength hypothesis
driver_points_before_race	results + races	Same-season recorded race points before current race.	Season strength; standings ablation
constructor_points_before_race	results + races	Points aggregated per constructor-race, then current race excluded.	Team strength; standings ablation
year	races	Calendar season, known before event.	Era drift hypothesis; context ablation
round	races	Calendar round, known before event.	Season progression; context ablation

Driver career metrics subtract the current outcome; recent rates shift by one race before rolling. Constructor outcomes are first aggregated at constructor-race grain, then excluded or shifted, preventing one teammate's current outcome from leaking into another teammate's features. Independent shifted-history recomputations and strictly earlier driver/constructor race-date assertions passed. [3, cell 100] [3, cell 101] [3, cell 102] [3, cell 104]

Interpretation limits. driver_prior_starts counts prior recorded entries, not verified physical race starts. Non-starters remain in the modeling population when present; there is no retrospective status filter silently redefining the target population. Constructor histories are also entry-based. year can absorb confounding and extrapolate poorly. Absolute Q1 seconds depend on circuit length, layout, weather and format; they are not normalized driver pace. grid=0 covers a special source convention and can include pit-lane starts or historical non-starters. Qualifying-record presence does not prove complete timing.

Season points are reconstructed from previous recorded main-race results rather than directly using official standings. Sprint points, historical scoring rules, penalties and later corrections can make them differ from as-of official standings. Earlier test-race outcomes can update form for a later test race in sequential pre-race evaluation; they never fit preprocessing or choose models, features, thresholds or hyperparameters. Full historical as-of data snapshots are unavailable.

9. Chronological split and training-only preprocessing

Split	Years	Rows	Podiums	Positive class (%)
Train	2006-2019	5775	804	13.92
Validation	2020-2021	780	117	15.00
Test	2022-2024	1359	204	15.01

Whole seasons and races remain in separate partitions: training 2006-2019, validation 2020-2021, test 2022-2024. Earlier records can supply history but do not become primary training examples. Assertions verify non-overlapping race IDs, chronological date boundaries, identical feature columns and aligned labels. Figure A27 shows class prevalence by split. [3, cell 108] [3, cell 114]

The reference preprocessor fits a median imputer and StandardScaler only on training data. Stored parameters transform validation/test and NN inference. RF uses training-fitted median imputation without scaling; LR and NN use their appropriate fitted scaling pipelines. Figures A28-A30 reveal distribution and scale changes rather than assuming scaling removes skew. No transformation is fitted on test observations. [3, cell 111] [3, cell 116]

10. Models, statistical baseline and training design

Model	Implemented configuration	Reason and principal limitation
Logistic Regression	Median imputation; StandardScaler; max_iter=2000; random_state=42.	Interpretable statistical baseline. Linear log-odds constrain interactions; correlated predictors complicate coefficients.
Random Forest	Median imputation; 200 trees; max_depth=10; min_samples_leaf=5; n_jobs=-1; random_state=42.	Nonlinear/interacting predictors. Higher train scores indicate capacity and possible overfitting; extrapolation is weak.
Shallow neural network	Dense 32 ReLU; dropout 0.2; Dense 16 ReLU; dropout 0.1; sigmoid output. Adam 0.001; binary cross-entropy.	Flexible shallow baseline. Development scores depend on validation early stopping and stochastic training.

The three models use the same rows and selected features; they constitute three meaningful modeling attempts. LR is the statistical baseline, not an unexecuted dummy model. The NN allows up to 100 epochs with batch size 64; early stopping monitors validation loss with patience 10 and restores the best weights. Version 13 trained for **35 epochs**, restoring **epoch 25** (best validation loss **0.23907172679901123**). The random seed is 42. Figure A31 shows the actual learning history. [3, cell 116] [3, cell 118] [3, cell 120]

Raw-data majority accuracy is only a class-imbalance illustration, not a separately trained/tested dummy baseline. RF's training AP 0.9019 versus validation AP 0.7315 is a substantial generalization gap. NN training loss reduction with flatter validation loss supports the need for early stopping; more capacity is not automatically better.

11. Metrics, model selection and held-out evaluation

AP, labeled PR-AUC in the comparison tables, is **average_precision_score**; it is not trapezoidal PR integration. Keras training-history PR-AUC is a different approximation and is not substituted for comparison-table AP. ROC-AUC measures global ranking separation. Precision measures the fraction of predicted podiums that are true; recall measures captured podiums; F1 balances the two at a chosen threshold. Accuracy is supplementary because non-podiums dominate.

Each model's threshold is selected for validation F1 over 0.05-0.95 in 0.01 increments. The model is then ranked by validation AP, with validation F1 as the tie-breaker. These choices occur before final test reporting. No test-based replacement is made. Figures A32-A33 compare validation metrics and precision-recall curves. [3, cell 122] [3, cell 126]

Train metrics

Model	ROC-AUC	AP	F1	Precision	Recall	Accuracy	Threshold
Logistic Regression	0.9351	0.7002	0.6853	0.5940	0.8097	0.8965	0.29
Random Forest	0.9805	0.9019	0.7998	0.7317	0.8818	0.9385	0.35
Neural Network	0.9437	0.7330	0.6875	0.5838	0.8358	0.8942	0.28

Validation metrics

Model	ROC-AUC	AP	F1	Precision	Recall	Accuracy	Threshold
Logistic Regression	0.9318	0.7363	0.7034	0.6975	0.7094	0.9103	0.29
Random Forest	0.9280	0.7315	0.6964	0.6615	0.7350	0.9038	0.35
Neural Network	0.9258	0.7496	0.7034	0.6975	0.7094	0.9103	0.28

Test metrics

Model	ROC-AUC	AP	F1	Precision	Recall	Accuracy	Threshold
Logistic Regression	0.9347	0.7063	0.6808	0.6532	0.7108	0.8999	0.29
Random Forest	0.9314	0.6655	0.6555	0.5735	0.7647	0.8793	0.35
Neural Network	0.9299	0.6847	0.6381	0.6204	0.6569	0.8882	0.28

The NN is selected with validation AP **0.7496** and threshold **0.28**; LR and RF thresholds are **0.29** and **0.35**. NN test AP is **0.6847**. Although LR test AP is higher (**0.7063**), it is not used to reverse model selection. All nine model/split rows have the required metrics, bounded values and their frozen validation-selected thresholds. [3, cell 123] [3, cell 128]

Development-data caveat. Validation data are reused for thresholds, early stopping and comparison, so validation performance is a development estimate. Earlier full-history EDA included later test years, and test results were inspected before subsequent Phase 8 analysis. The notebook discloses this history. The later validation experiments are exploratory, not newly untouched-test-confirmed improvements.

12. Two preprocessing-order experiments

Both experiments preserve the chronological training/validation rows, all 16 features and the same Logistic Regression (max_iter=2000, random_state=42). Every learned transformation is fitted on training data only. Comparisons use validation ROC-AUC, AP and validation-selected F1. Neither experiment changes the frozen Phase 7 model or thresholds. Figures A34 and A36-A37 show the executed results and transformed distributions. [3, cell 132] [3, cell 137]

12.1 Median imputation versus standard scaling order

Configuration	ROC-AUC	AP	F1	Threshold
B: StandardScale → Impute	0.9319	0.7371	0.7034	0.29
A: Impute → StandardScale	0.9318	0.7363	0.7034	0.29
C: Impute → RobustScale	0.9288	0.7311	0.7059	0.43

For a fixed increasing affine transform, the median of transformed observations equals the transformed median. However, fitting StandardScaler after median imputation uses a different empirical mean/variance from fitting it while ignoring missing entries. Thus the complete fitted pipelines need not commute. Maximum validation transformed-value and probability differences are **0.028441** and **0.002941**. They are not identical at a 1e-6 probability tolerance. Configuration C changes the scaling rule and is a reference comparison, not a second order reversal.

12.2 Quantile capping versus median imputation order

Configuration	ROC-AUC	AP	F1	Threshold
D: Impute → Cap → Scale	0.931327	0.726222	0.702703	0.43
E: Cap → Impute → Scale	0.931327	0.726222	0.702703	0.43

Training-fitted 1st/99th-percentile capping is applied to Q1 duration, driver/constructor prior exposure and driver/constructor pre-race points. D imputes, caps and scales; E caps, imputes and scales. Imputed mass can alter fitted quantiles. The maximum transformed difference is **0.004359**, and the maximum predicted-probability difference is about **0.000012**. Rounded ranking/F1 metrics agree, but the transformations/probabilities are not identical at the specified tolerance. Both AP values (0.726222) are lower than reference A (0.7363): no artificial gain is claimed.

13. Feature-group ablation

The LR reference is retrained after removing each of six groups while retaining the same training/validation rows and median-imputation/standard-scaling setup. The displayed AP drop is reference AP minus ablated AP; a negative value means the ablated model scored slightly better. Figure A35 shows the contribution comparison. [3, cell 134]

Configuration	Features	ROC-AUC	AP	F1	AP drop
Without grid	14	0.9295	0.7239	0.7089	0.0123
Without standings	14	0.9315	0.7252	0.7059	0.0110
Without qualifying	12	0.9240	0.7260	0.7054	0.0102
Without race_context	14	0.9318	0.7339	0.7078	0.0024
Full model	16	0.9318	0.7363	0.7034	0.0000
Without driver_form	13	0.9308	0.7363	0.7102	-0.0001
Without constructor_form	13	0.9320	0.7393	0.7029	-0.0031

Removing grid, reconstructed standings and qualifying yields the largest AP losses: **0.0123, 0.0110 and 0.0102**. Removing constructor form slightly improves validation AP by about **0.0031**; removing driver form has little effect. Correlated information can make a useful feature group appear redundant. These findings are specific to LR and this validation period, are rounded at display precision, and do not establish causal feature effects or NN feature contributions. No feature is removed on this basis.

14. Global and local explainability

All XAI here explains the fitted **Logistic Regression baseline**, not the selected NN. Permutation importance measures the validation AP loss after a feature is shuffled (five repeats, random_state=42). It ranks qualifying position first (**0.1219**) and grid second (**0.1128**). SHAP uses a LinearExplainer with a training background sample of at most 200 and a validation sample of at most 100; grid and qualifying position have the largest mean absolute log-odds attributions (**0.8851** and **0.8698**). Figures A38-A40 show the two global perspectives. [3, cell 140] [3, cell 141] [3, cell 142]

Permutation sensitivity and SHAP attribution need not rank correlated predictors identically. For example, year has substantial SHAP attribution but small negative permutation AP importance in this validation sample. SHAP values are in log-odds, not probability percentage points. Neither method proves a causal contribution to a podium.

The local example is **driverId 1, raceId 1037**, with LR predicted podium probability **0.8940** and actual podium label 1. A SHAP waterfall (Figure A41) and an actual plotted LIME explanation (Figure A42) use the

same validation record. LIME uses 1,500 perturbations, categorical binary availability flags, no continuous discretization, and the fitted LR preprocessing/classifier. Its local weighted surrogate R-squared is **0.8082**. [3, cell 143]

LIME and local SHAP share **grid and qualifying position** in their top five absolute attributions. Higher numerical grid/qualifying positions have negative LIME response weights, while this record's favorable observed positions have positive SHAP contributions relative to its background. A local slope/response weight and displacement from a reference answer different questions; signs and units need not match. LIME's availability-flag effects can reflect unrealistic independent perturbations of correlated features. Its fit is approximate, not a causal or global faithfulness guarantee.

Noncausal interpretation: permutation importance, SHAP and LIME describe how the fitted model relies on its inputs. They do not establish that changing a feature will change the sporting outcome, and they do not identify intervention effects.

15. Complete and partially missing inference

predict_podium accepts a dictionary of 16 engineered pre-race inputs, rejects unknown names, converts numerical values, reconstructs availability flags, and applies the stored training-fitted preprocessor. Year and round are required. It returns an NN podium probability, thresholded class and missing-feature list. It deliberately remains NN-only rather than dynamically dispatching through a model-name variable; the validation-selected model happens to be the NN in Version 13. [3, cell 145]

Demonstration	NN probability	Class	Threshold	Missing features
Complete validation record	0.7233	1	0.28	None
Partially missing record	0.4717	1	0.28	q1_seconds; driver_last5_podium_rate; constructor_last5_podium_rate

Both demonstrations executed successfully. They need not be the same original validation record, so their probability difference is not an isolated missingness effect. This is feature-level inference: it does not fetch race-weekend information or automatically construct historical form from IDs. Missingness is handled using training statistics rather than by fitting on the new record. [3, cell 146]

16. Limitations, reproducibility and conclusion

Historical and measurement limits. Shared-drive deduplication sacrifices detail. Entry counts include some non-starters. Qualifying timing/coverage and sporting formats drift across eras. Grid zero is a source convention, and absolute Q1 times are not circuit-normalized pace. Mechanical status labels and eligible-start proxies are imperfect; constructor rebranding changes identity boundaries. The Q1 anomaly's historical cause remains unknown.

Statistical and modeling limits. Home-country adjustment uses grid bands rather than a fully specified driver/team/era model. The race-cluster interval does not fully address repeated-driver dependence. Validation is reused, model capacities generalize differently, calibration is not established, and probabilities are not guaranteed outcomes. Earlier test inspection and full-history EDA prevent a claim of a pristine untouched-test study. No causal interpretation is warranted for associations or XAI.

Reproducibility evidence. Final Version 13, scriptVersionId 359134370, succeeded in 127.6 seconds. Its 151 cells comprise 125 Python and 26 Markdown cells; all Python execution counters run 1-125 without an error cell. All 42 output figures were loaded and retrieved. Complete PK/FK assertions, unique driver-race grain, temporal provenance, raw CSV fingerprints, model/split metric checks, preprocessing/ablation checks, XAI dimensions and inference checks passed. The final export cell reports 23 report tables. No training code, hyperparameters, dataset or split was modified for this report. [3, cell 149] [3, cell 150]

Conclusion. The project provides an executed and auditable driver-entry podium prediction pipeline. Grid, qualifying and reconstructed season points provide important predictive information in the implemented comparisons. NN remains the validation-selected model at threshold 0.28 despite LR's higher observed test AP.

The report preserves this distinction, includes the actual uncertainty and sensitivity evidence, and treats model explanations as reliance rather than causation.

References and provenance

[1] **MS1 Description (Project).pdf**, supplied milestone specification, 8 pages. Governs the prediction cutoff, data-engineering questions, modeling/evaluation, experiments, XAI and deliverables.

[2] **Introduction for Formula 1 Beginners (Project).pdf**, supplied contextual guide, 7 pages. Provides sporting definitions, historical oddities and coverage considerations; not used as a numerical results source.

[3] **Acl_Team_61, final saved Kaggle Version 13**, scriptVersionId 359134370. Successful execution and output images inspected; 1-based cell locators used throughout. Open the exact saved version.

The Version 13 sharing dialog shows Public selected and provides the viewer URL. This report records the existing setting; it neither changes sharing nor submits the assignment. An independent signed-out evaluator-access test was not performed. An executed notebook and report are not substitutes for the other submission requirements in [1].

Appendix A. Complete notebook figure catalogue

All 42 original figures below are copied from the saved Version 13 output files. Composite multi-panel outputs count as one figure, as in the notebook. No plot was regenerated or numerically reconstructed for this report. Source cell references identify the exact original location.

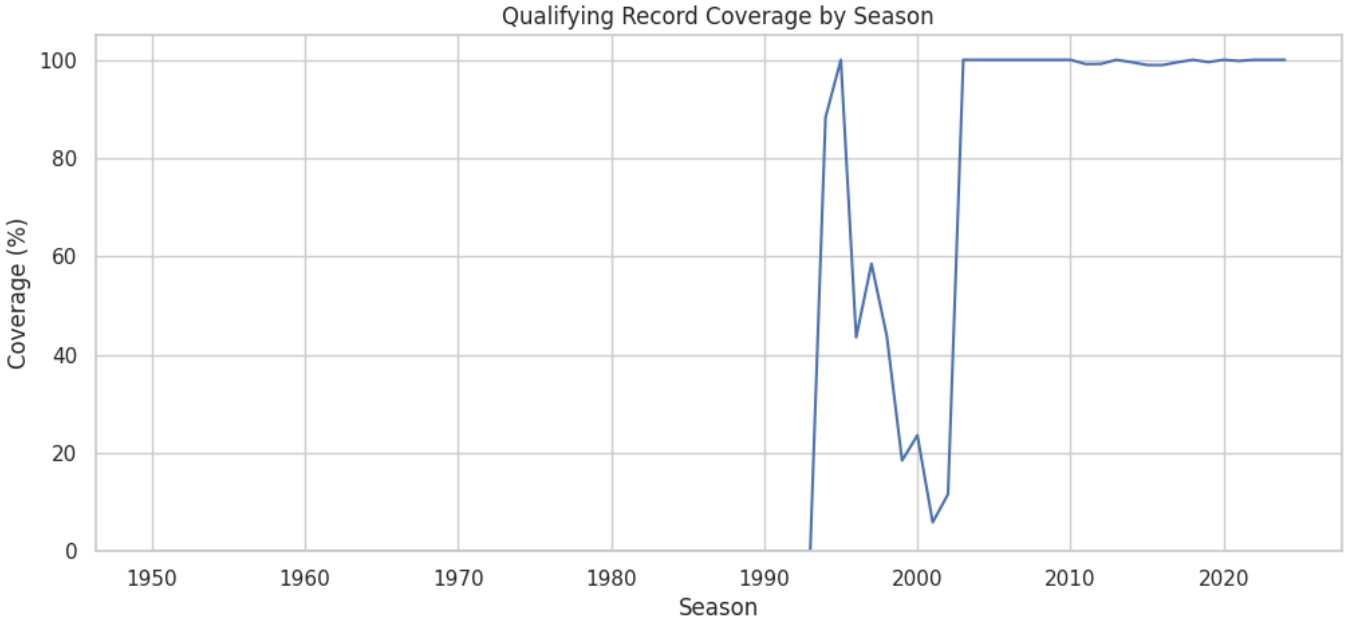


Figure A1. Qualifying coverage audit. Initial qualifying-record coverage across seasons reveals substantial historical coverage drift; it is not a measure of all session timing completeness. [3, cell 13] Original output: __results___12_0.png.

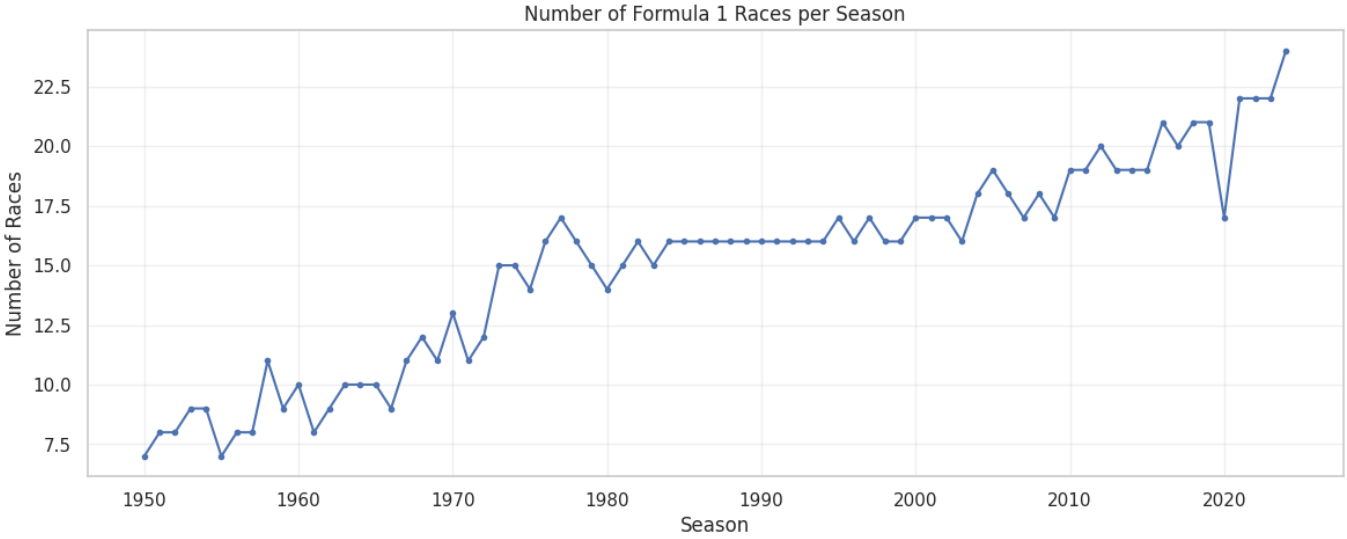


Figure A2. Championship races per season. Growing and unequal season lengths make raw historical performance totals difficult to compare directly. [3, cell 17] Original output: __results___16_0.png.

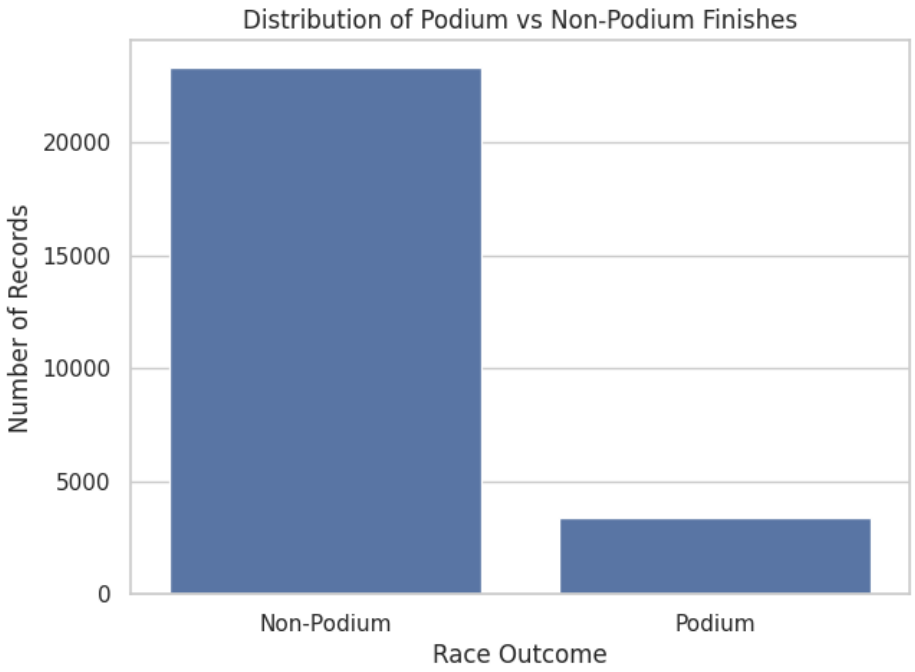


Figure A3. Raw target distribution. The raw podium class is rare; high majority-class accuracy can conceal failure to detect any podiums. [3, cell 18] Original output: __results___17_1.png.

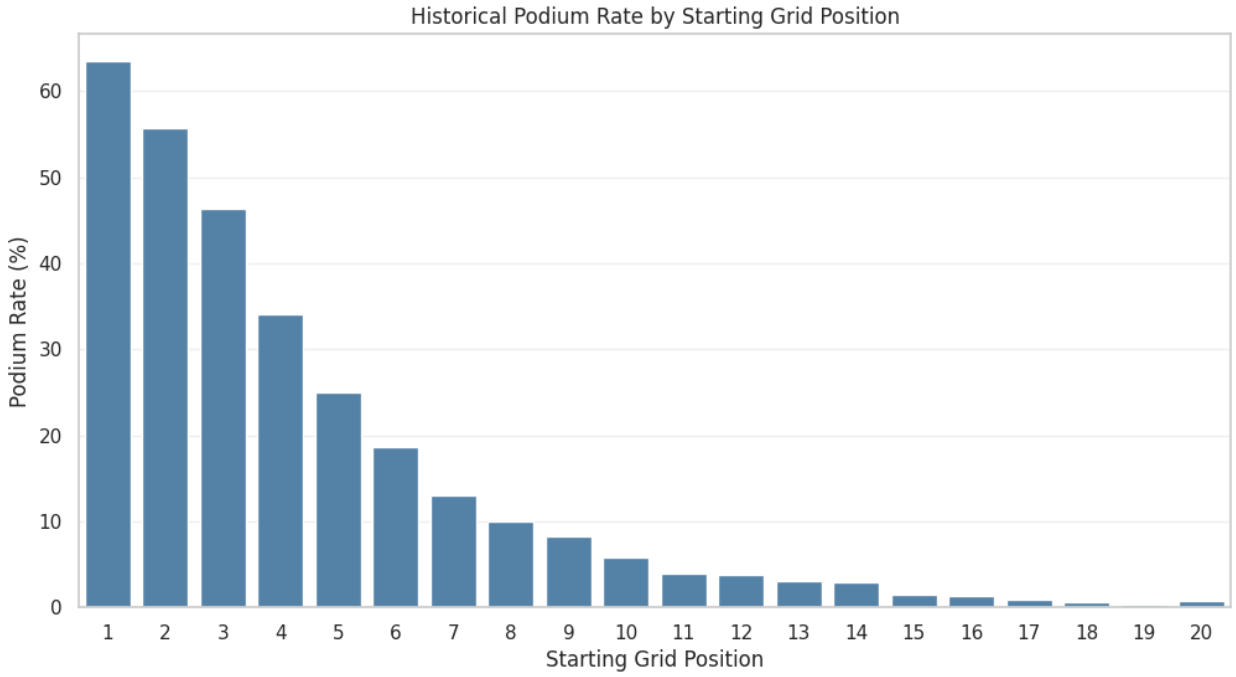


Figure A4. Historical podium rate by grid. Positive grid positions show a strong historical association with podium frequency; raw rows and confounding limit causal interpretation. [3, cell 22] Original output: __results___21_0.png.

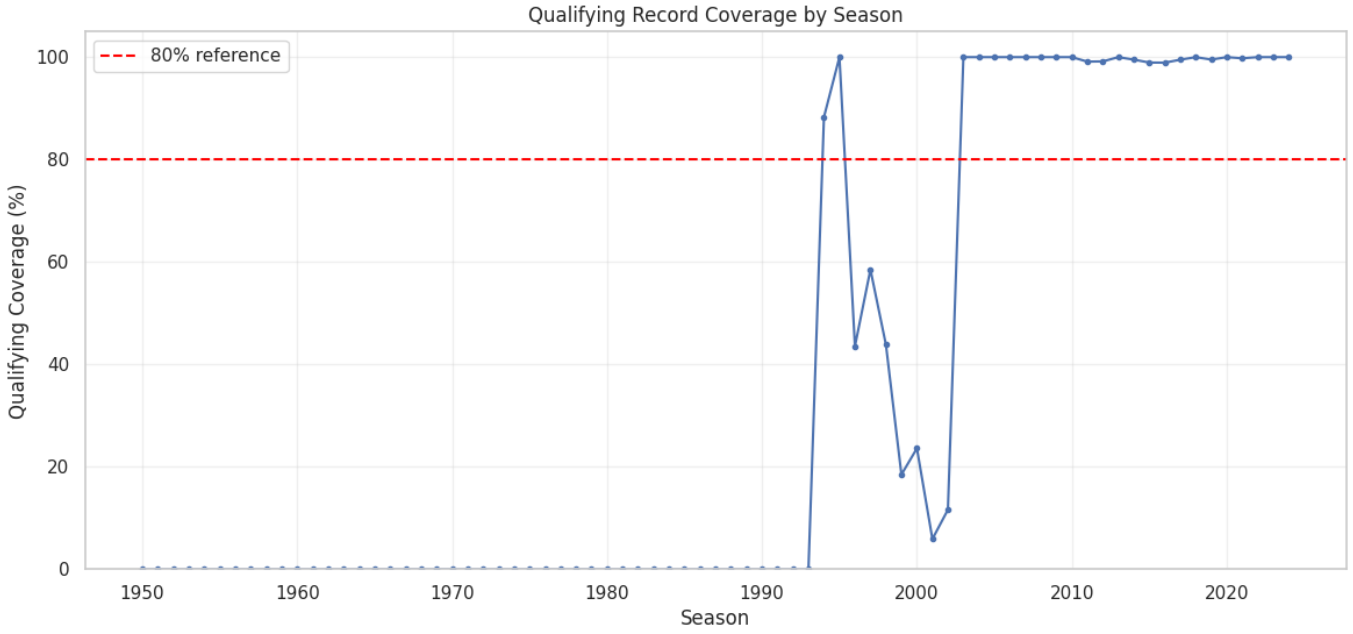


Figure A5. Qualifying coverage by season. The record-level coverage view motivates missingness indicators and the fixed modern training window. [3, cell 26] Original output: __results___25_0.png.

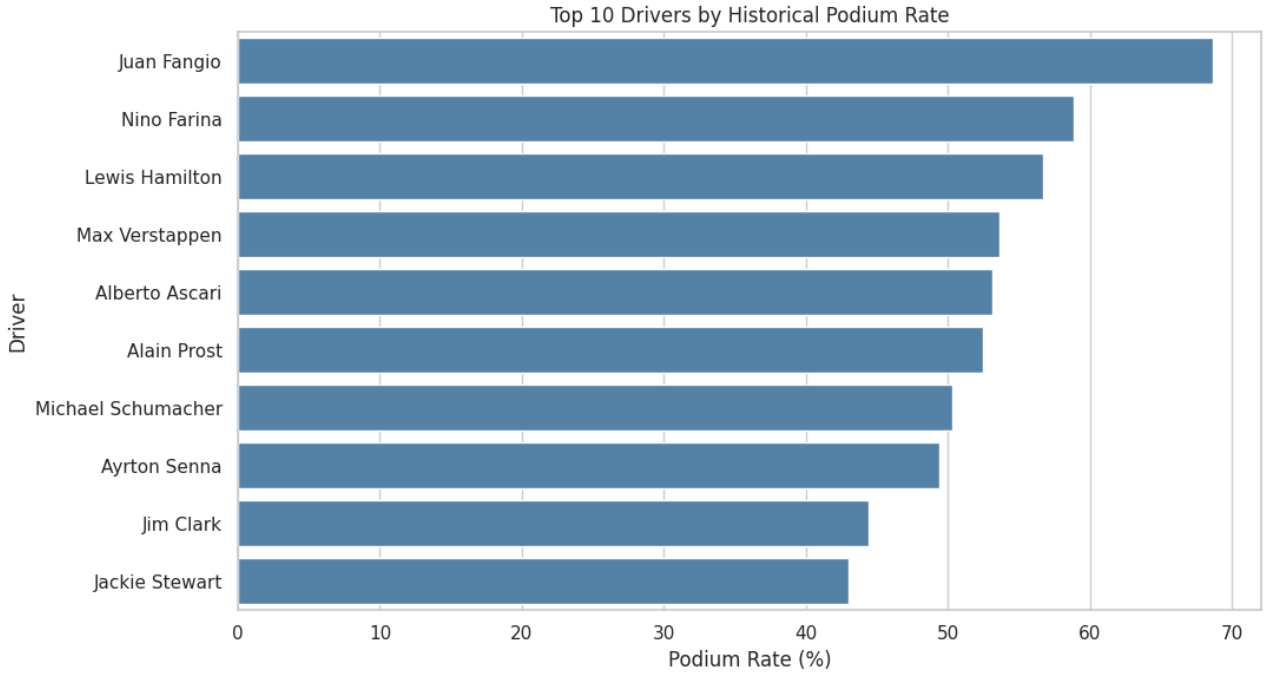


Figure A6. Driver historical podium rates. The minimum-exposure filter reduces very small-sample rankings, but raw shared-drive and numerator/denominator conventions still limit skill interpretation. [3, cell 31] Original output: __results___30_0.png.

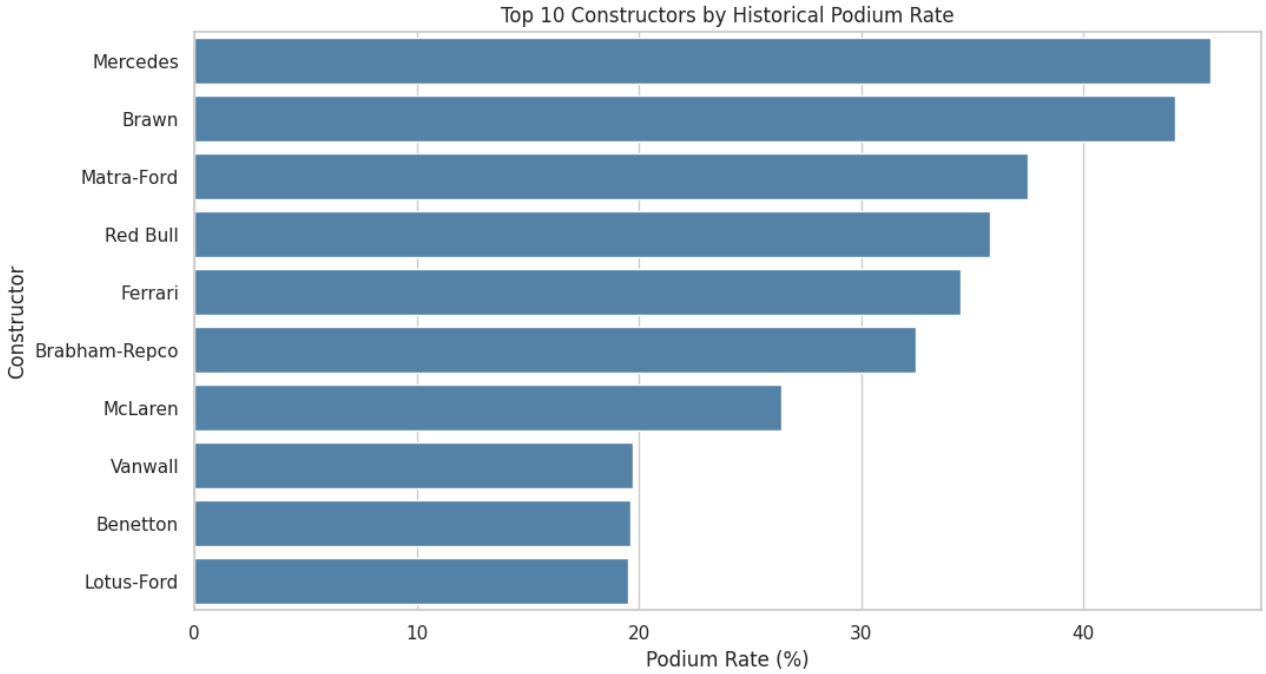


Figure A7. Constructor historical podium rates. Observed team differences motivate constructor-history features; era composition and rebranding remain confounders. [3, cell 33] Original output: __results___32_0.png.

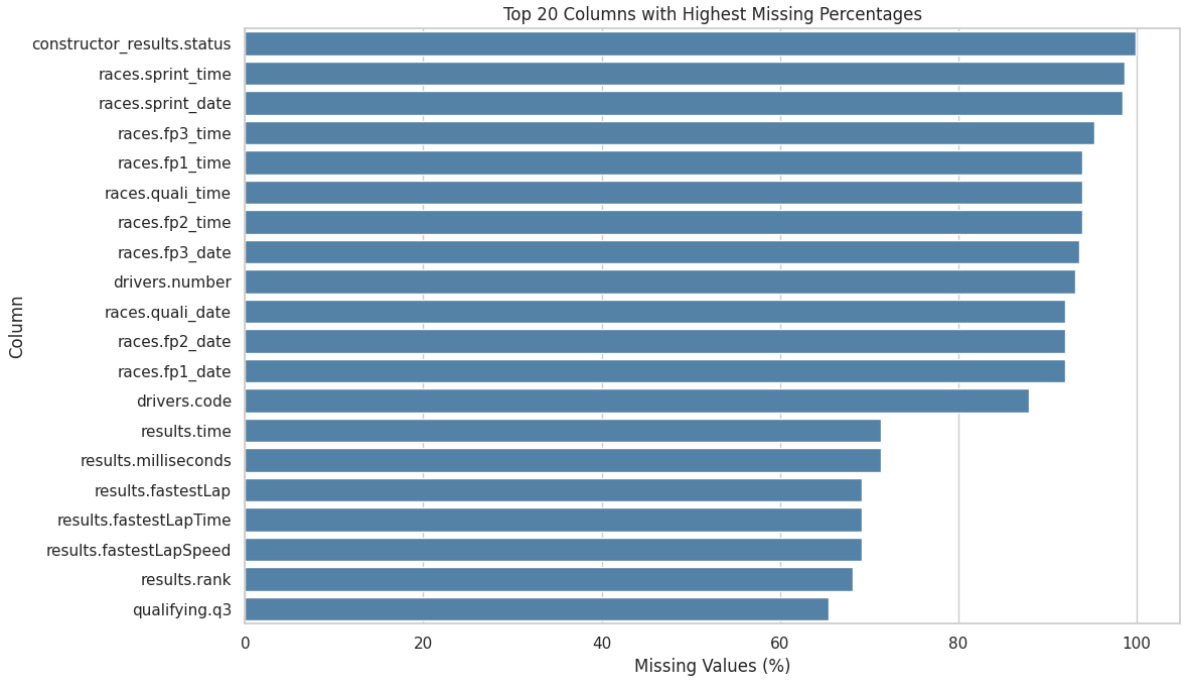


Figure A8. Columns with highest missingness. High missingness is concentrated in fields with structural or historical recording limitations; blanket row deletion would alter the population. [3, cell 36] Original output: __results___35_0.png.

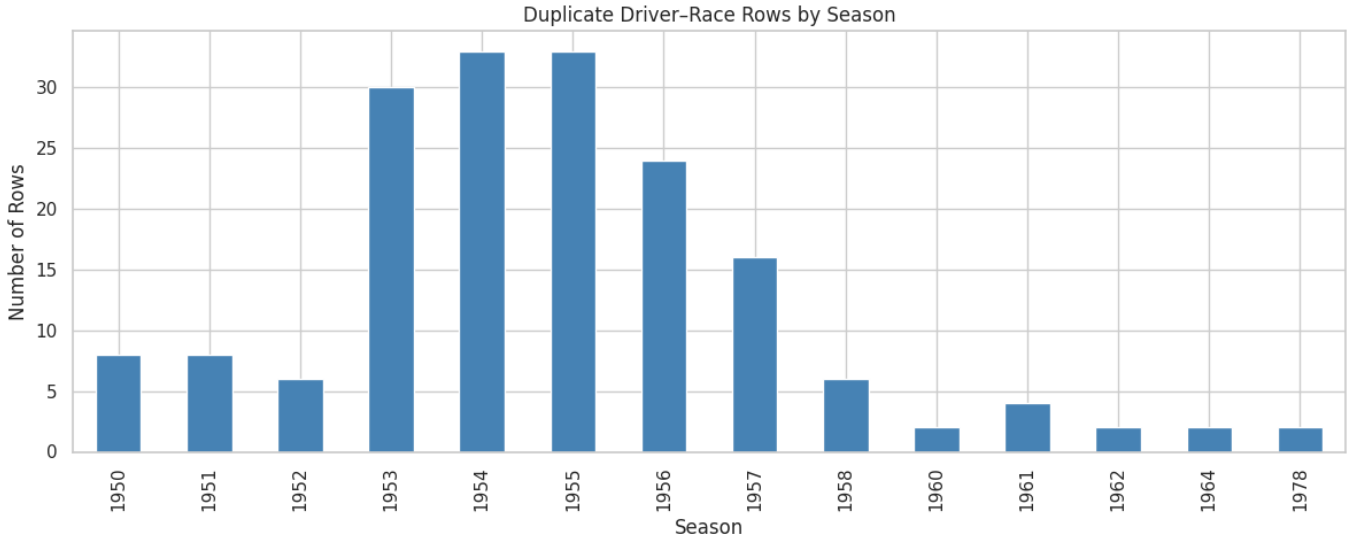


Figure A9. Repeated driver-race rows by season. Historical duplicate patterns support explicit shared-drive handling rather than treating every duplicate as an arbitrary data-entry mistake. [3, cell 38] Original output: __results___37_0.png.

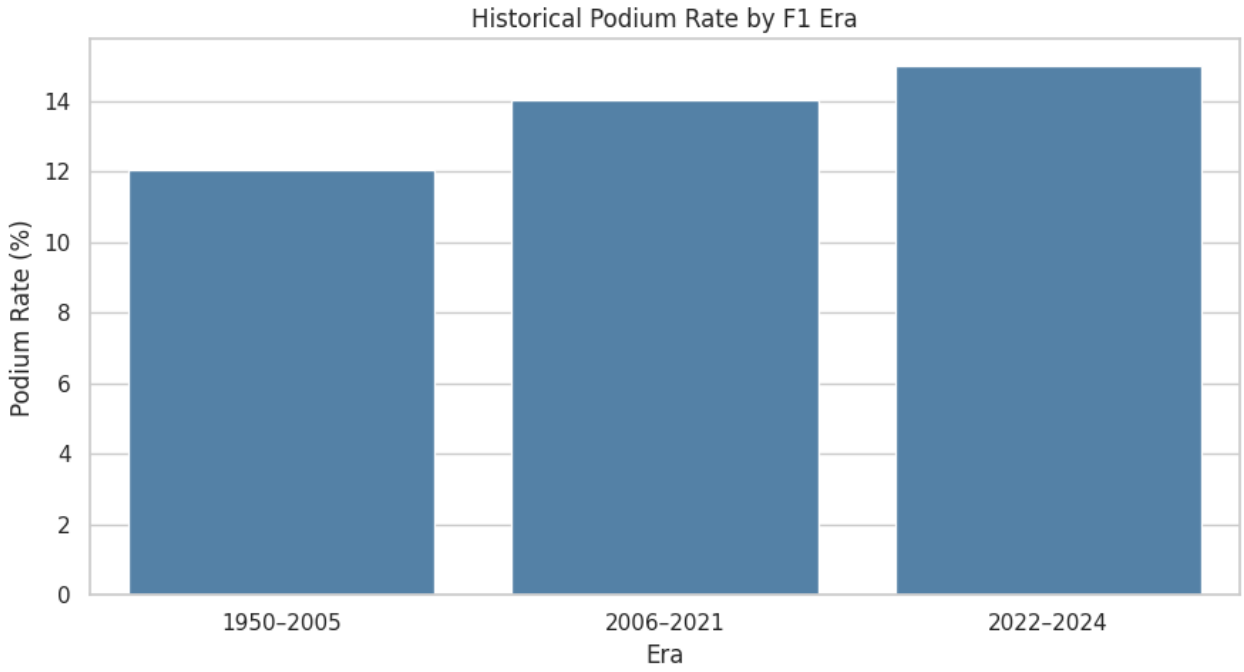


Figure A10. Podium prevalence by historical era. Different field sizes and participation regimes change the baseline positive rate across eras. [3, cell 40] Original output: __results___39_1.png.

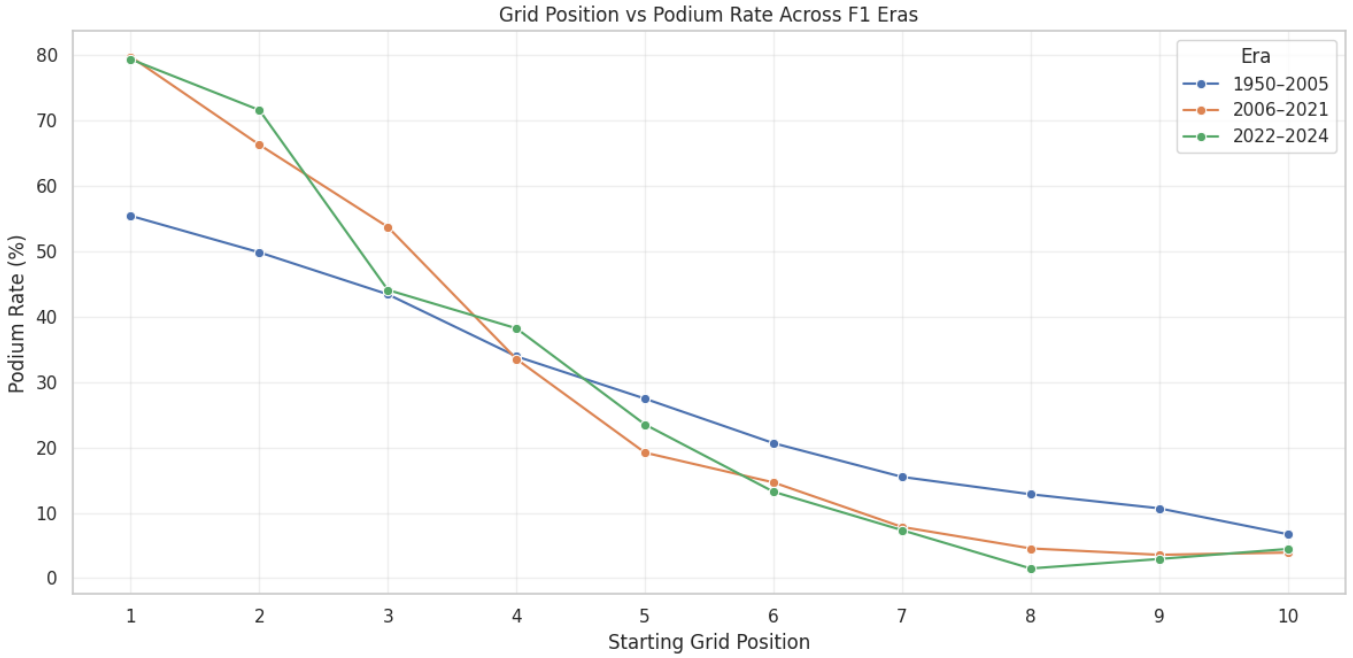


Figure A11. Grid association across eras. The grid-podium relationship differs across sporting contexts; pooled historical associations need era-aware interpretation. [3, cell 41] Original output: __results___40_0.png.

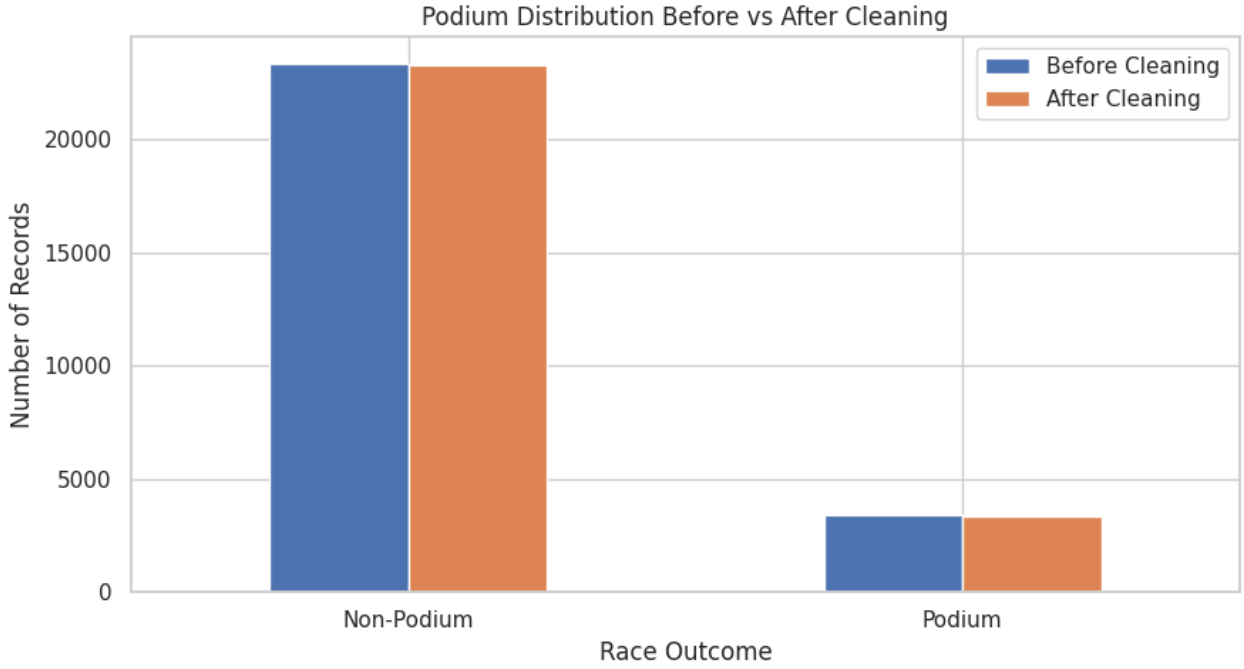


Figure A12. Podium distribution before and after deduplication. The deterministic grain correction changes counts while the printed podium prevalence changes only slightly, from 12.69% to 12.67%. [3, cell 54] Original output: __results___53_0.png.

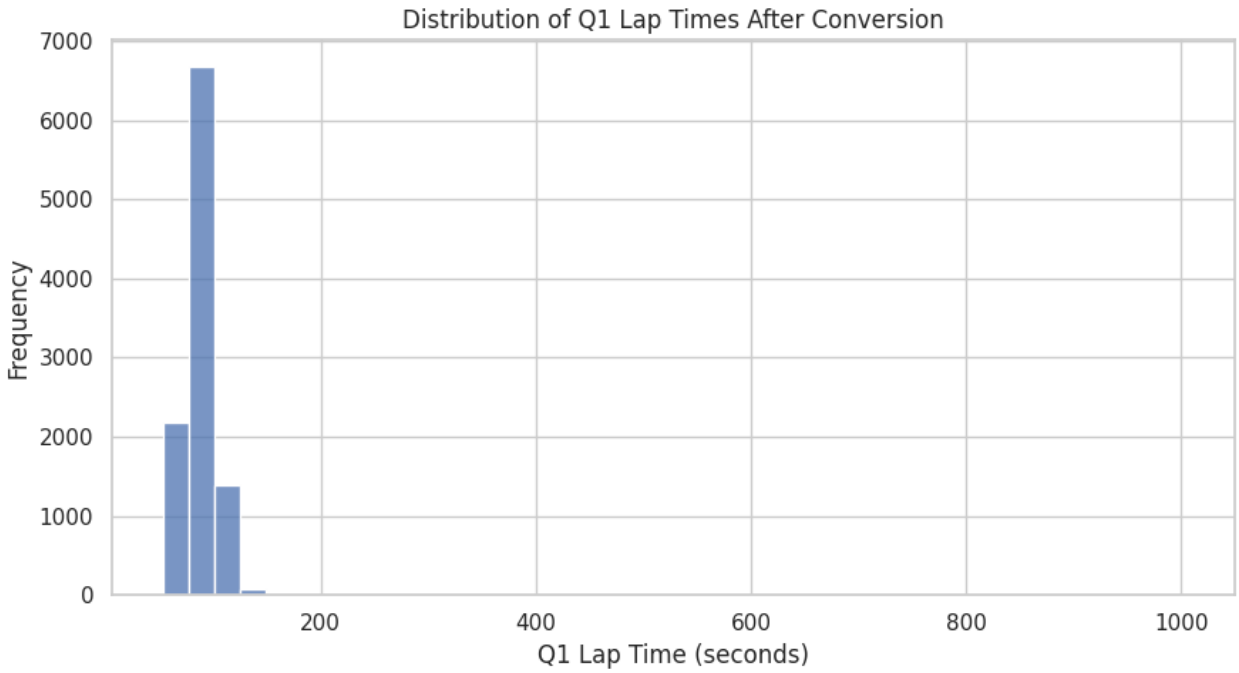


Figure A13. Converted Q1 duration distribution. Semantic conversion exposes the extreme duration without inventing a replacement value. [3, cell 55] Original output: __results___54_0.png.

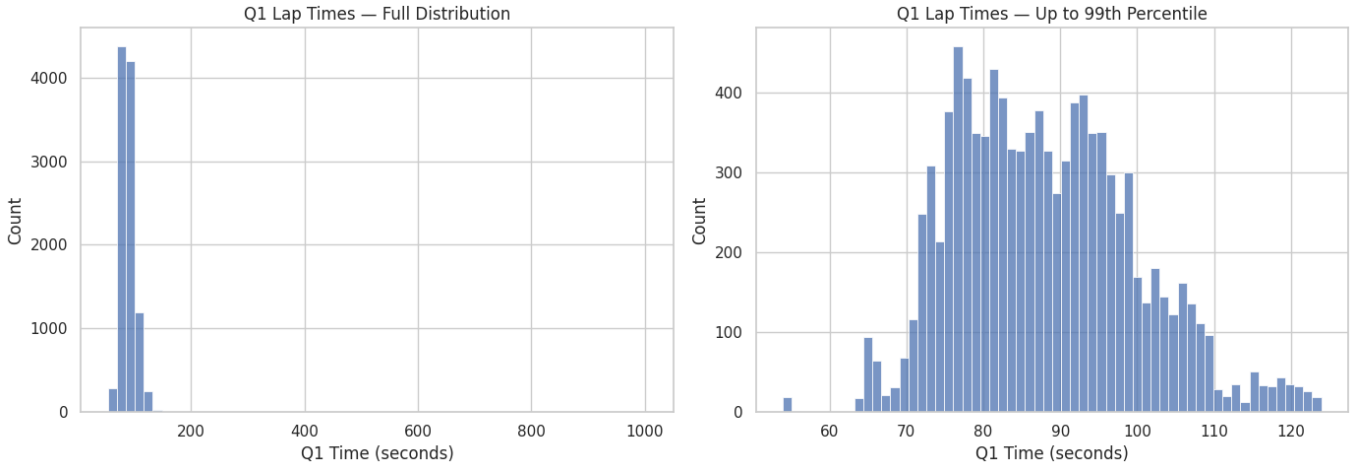


Figure A14. Q1 full versus trimmed distribution. The full-scale outlier obscures typical timing structure; the 99th-percentile panel is an exploratory view, not a deletion rule. [3, cell 60] Original output: __results___59_0.png.

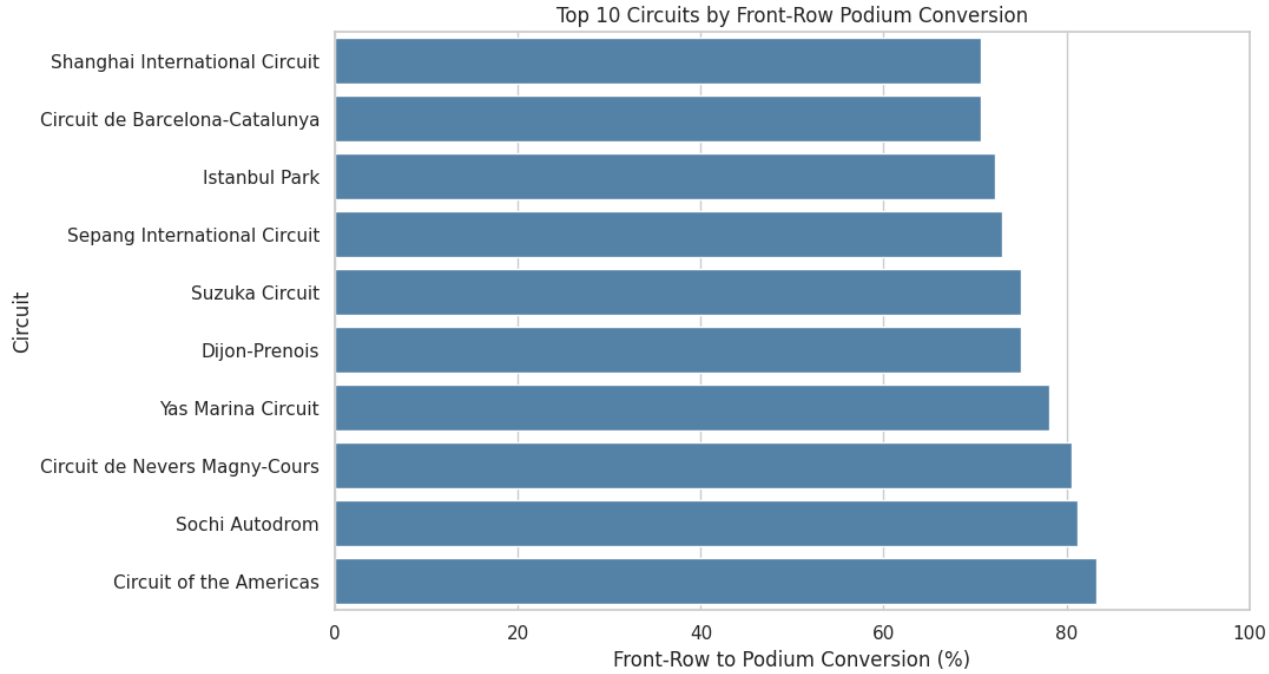


Figure A15. Circuit front-row conversion ranking. Eligible circuits require at least 10 front-row observations; COTA leads the displayed overall ranking. [3, cell 68] Original output: __results___67_0.png.

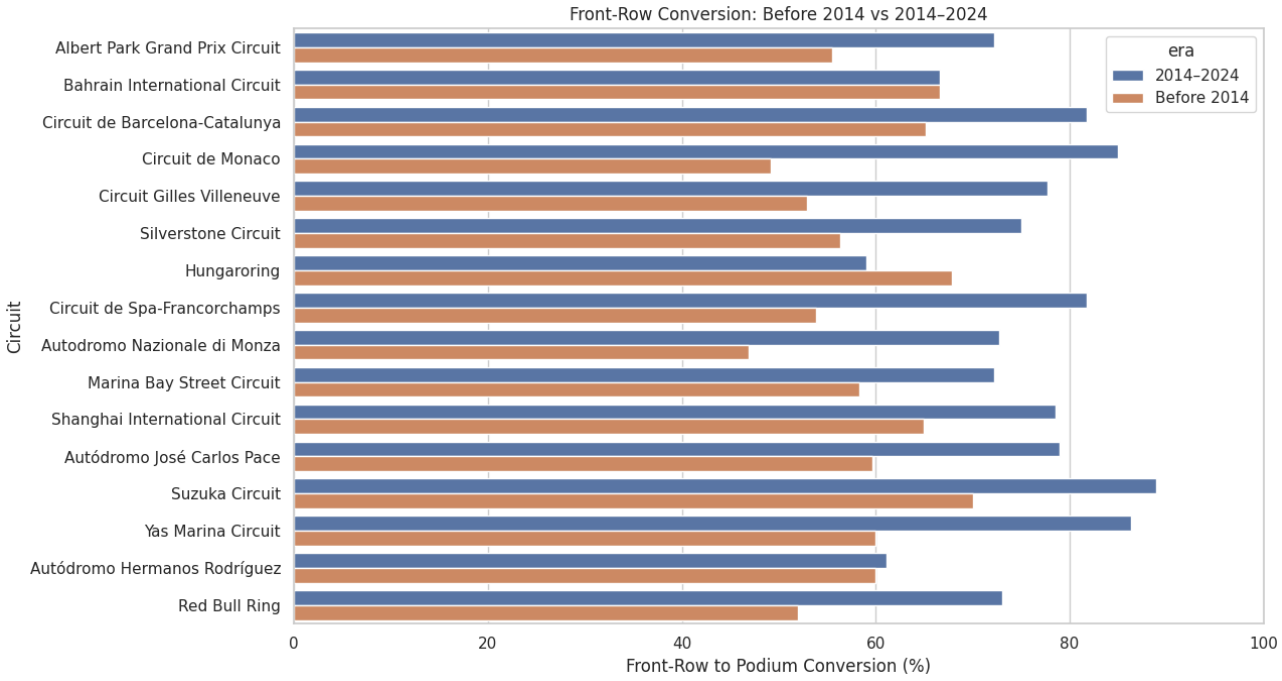


Figure A16. Circuit conversion before and after 2014. Only circuits with at least 10 observations in each era are compared; visible changes are descriptive. [3, cell 71] Original output: __results___70_0.png.

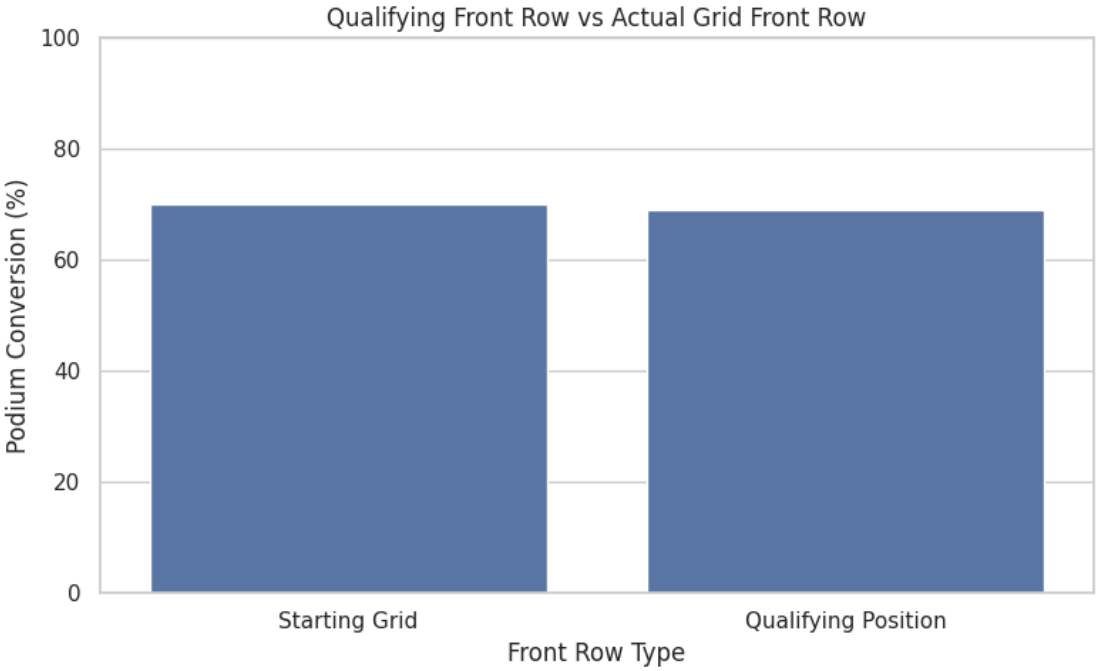


Figure A17. Qualifying front row versus actual grid front row. The comparable subset has different conversion under qualifying versus actual starting-grid definitions; the groups need not contain identical entries. [3, cell 73] Original output: __results___72_1.png.

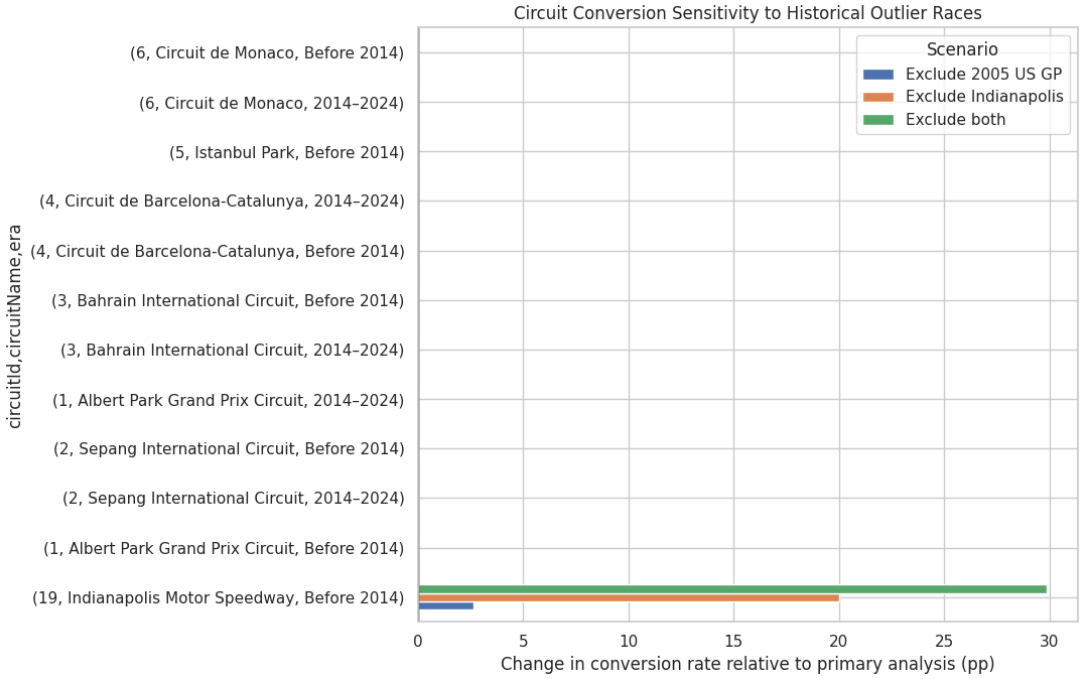


Figure A18. Circuit sensitivity to historical outlier races. Indianapolis and the unusual 2005 US GP are excluded separately and jointly to expose rate sensitivity and sample dependence. [3, cell 73] Original output: __results___72_6.png.

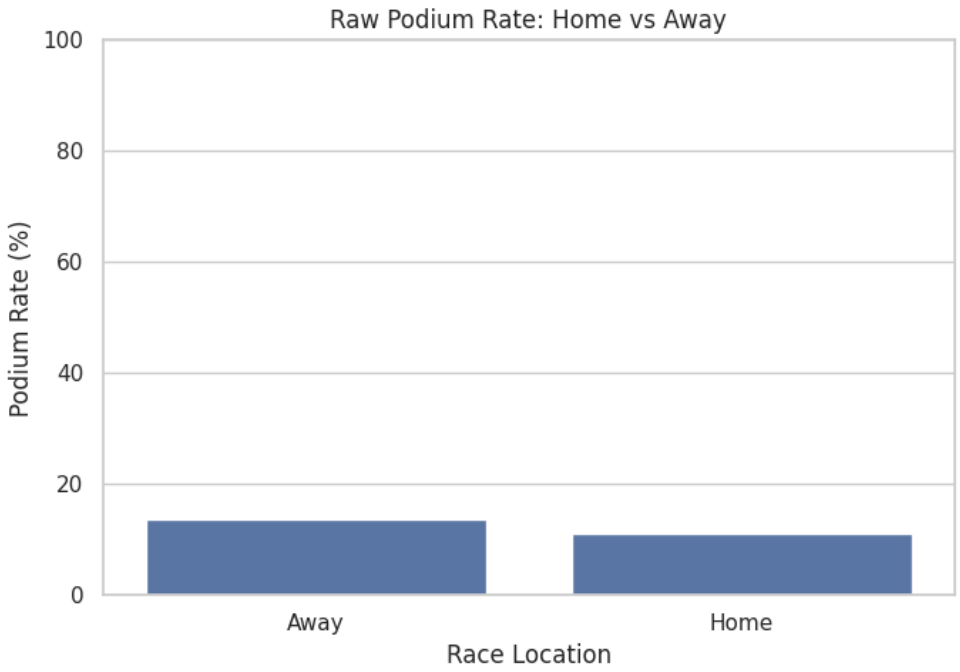


Figure A19. Unadjusted home versus away rate. Raw home and away averages differ in grid composition and cannot answer the controlled home-effect question alone. [3, cell 79] Original output: __results___78_1.png.

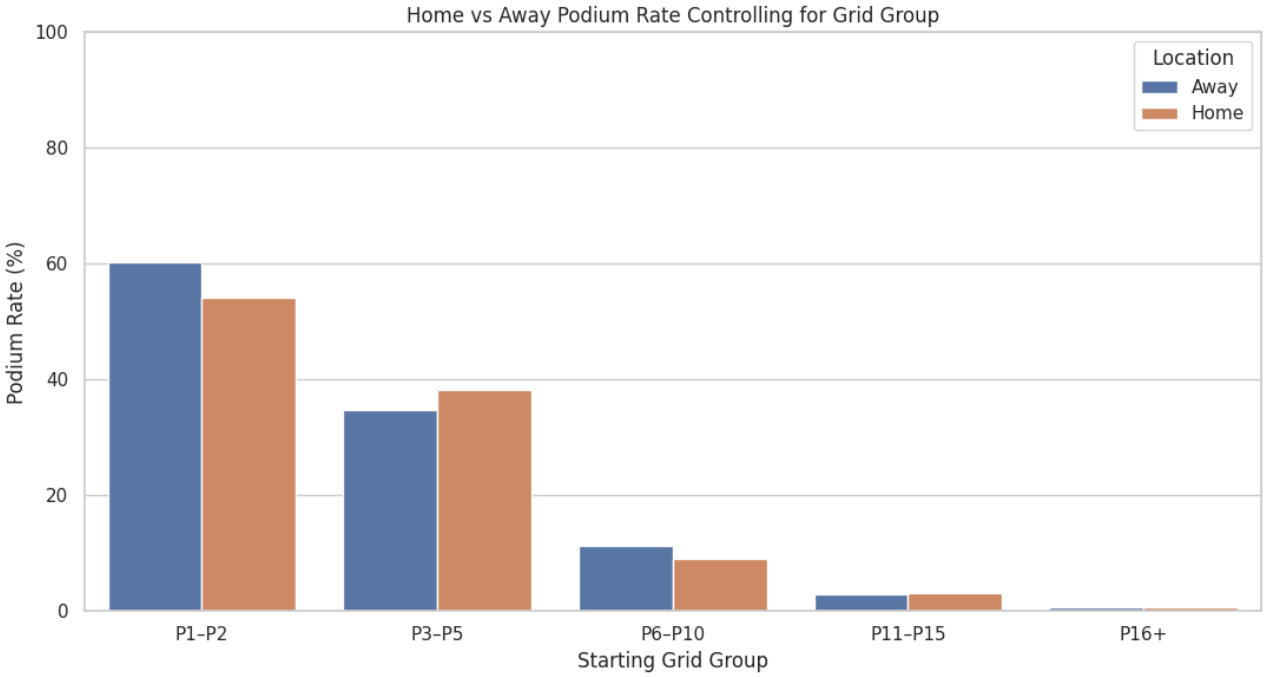


Figure A20. Home versus away within grid bands. Within-band comparisons and pooled standardization reduce grid composition differences but leave residual confounding. [3, cell 80] Original output: __results___79_1.png.

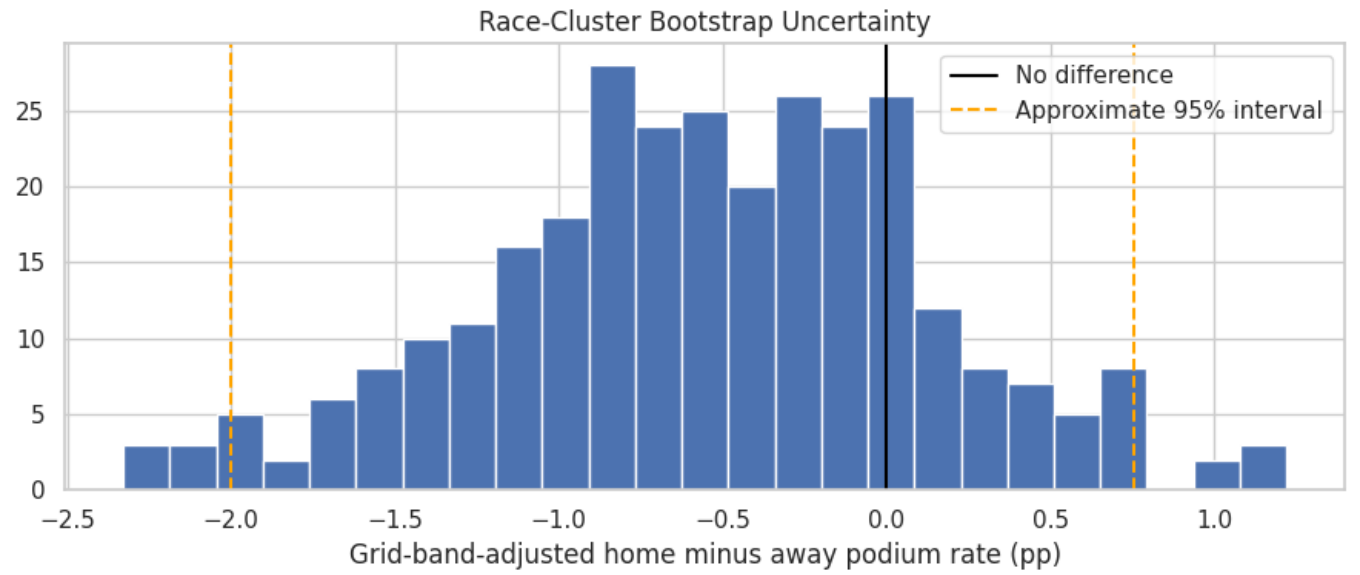


Figure A21. Home-effect race-cluster uncertainty. The approximate bootstrap interval includes zero; the small adjusted negative difference is not an established causal disadvantage. [3, cell 82] Original output: __results___81_4.png.

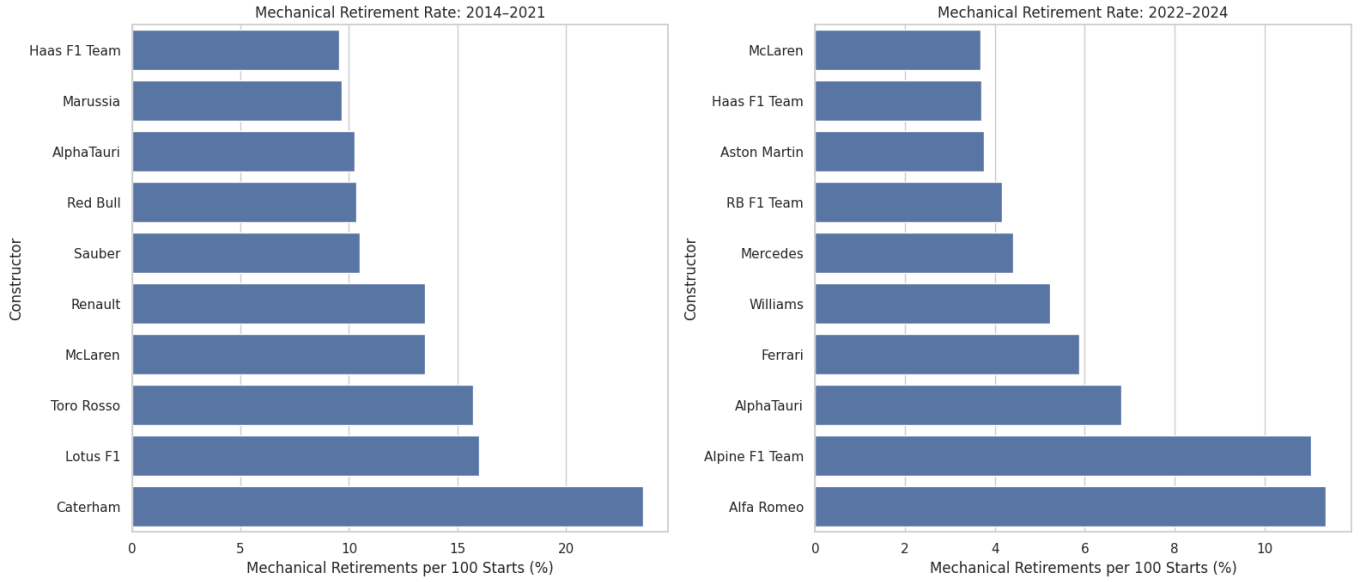


Figure A22. Broad mechanical-retirement ranking by era. The original status-based constructor rankings use an eligible-record start proxy and require minimum exposure. [3, cell 90] Original output: __results___89_0.png.

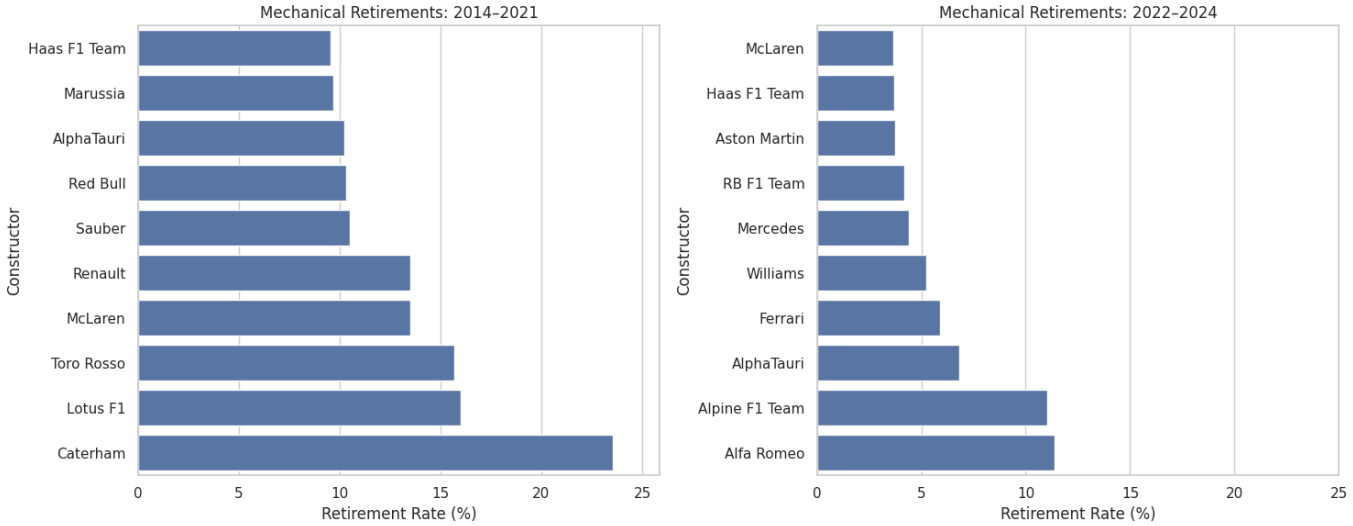


Figure A23. Final broad mechanical-retirement comparison. The final broad mapping yields lower overall later-era rates; classification and denominator assumptions accompany these values. [3, cell 95] Original output: __results___94_0.png.

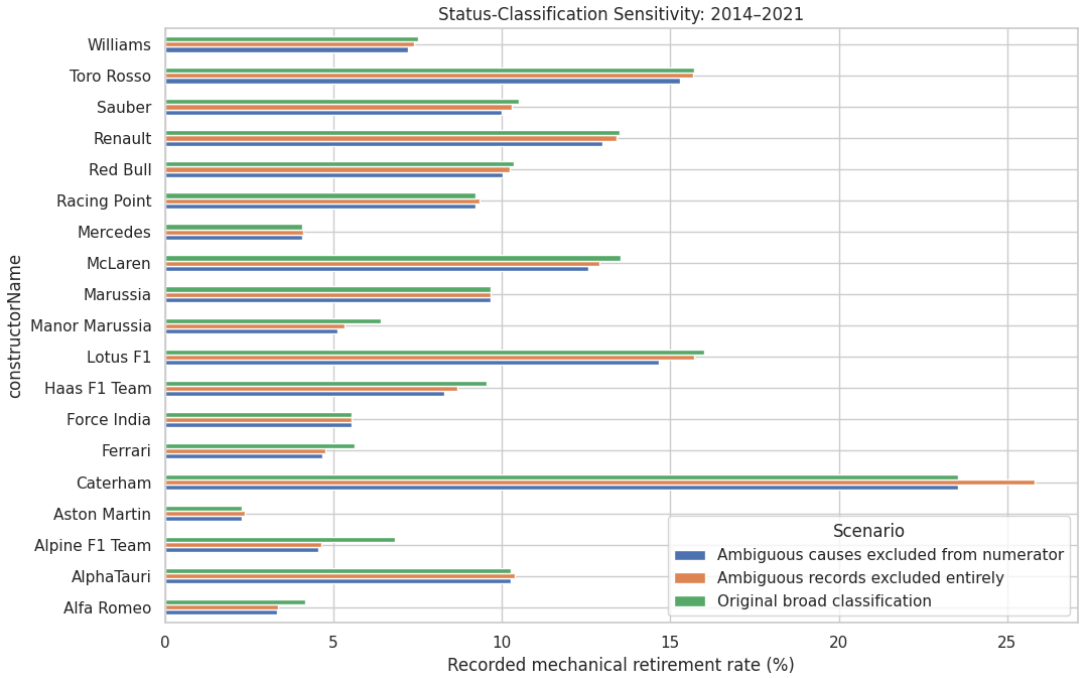


Figure A24. Status sensitivity, 2014-2021. Conservative ambiguous-status treatments alter some rates; Caterham remains the earlier-era leader. [3, cell 95] Original output: __results___94_3.png.

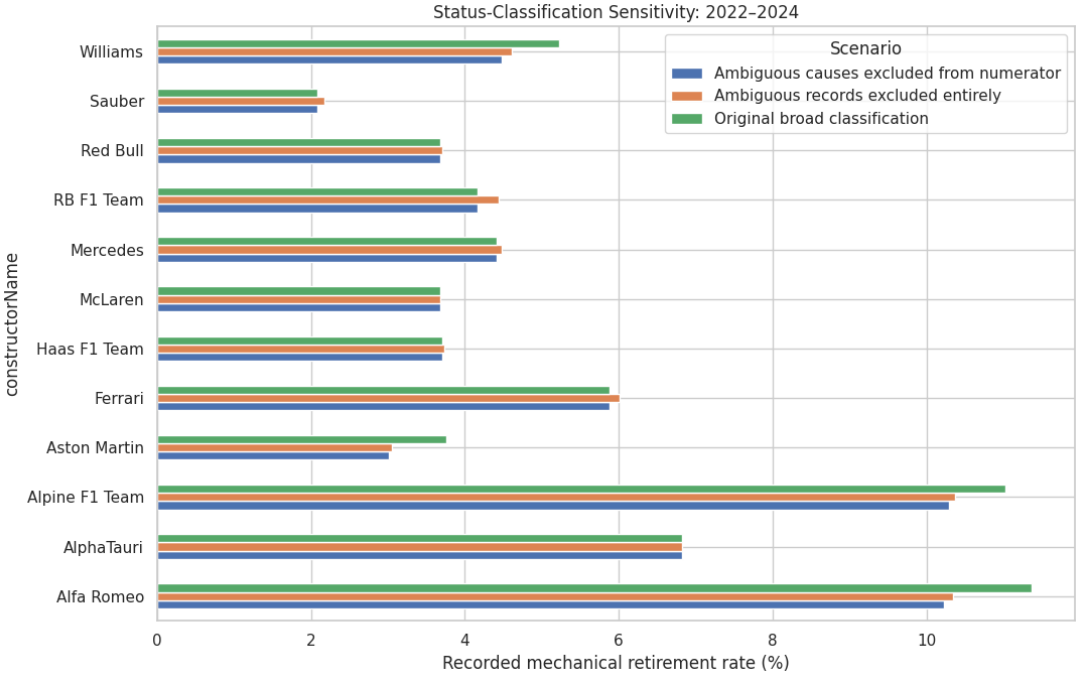


Figure A25. Status sensitivity, 2022-2024. Alpine overtakes Alfa Romeo under conservative ambiguous-status treatment, showing rank dependence on coding choices. [3, cell 95] Original output: __results___94_4.png.

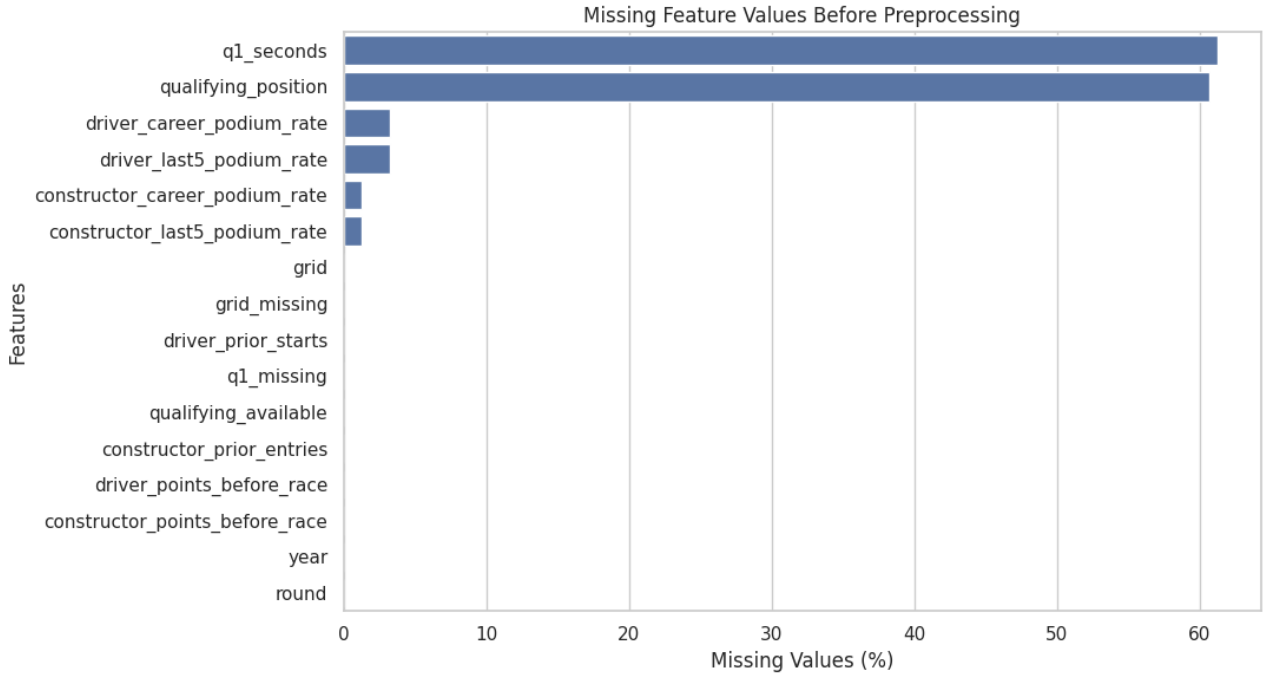


Figure A26. Model-feature missingness. Historical feature availability varies; availability flags and training-only imputation preserve entries without assuming missing values are ordinary observations. [3, cell 105] Original output: __results___104_0.png.

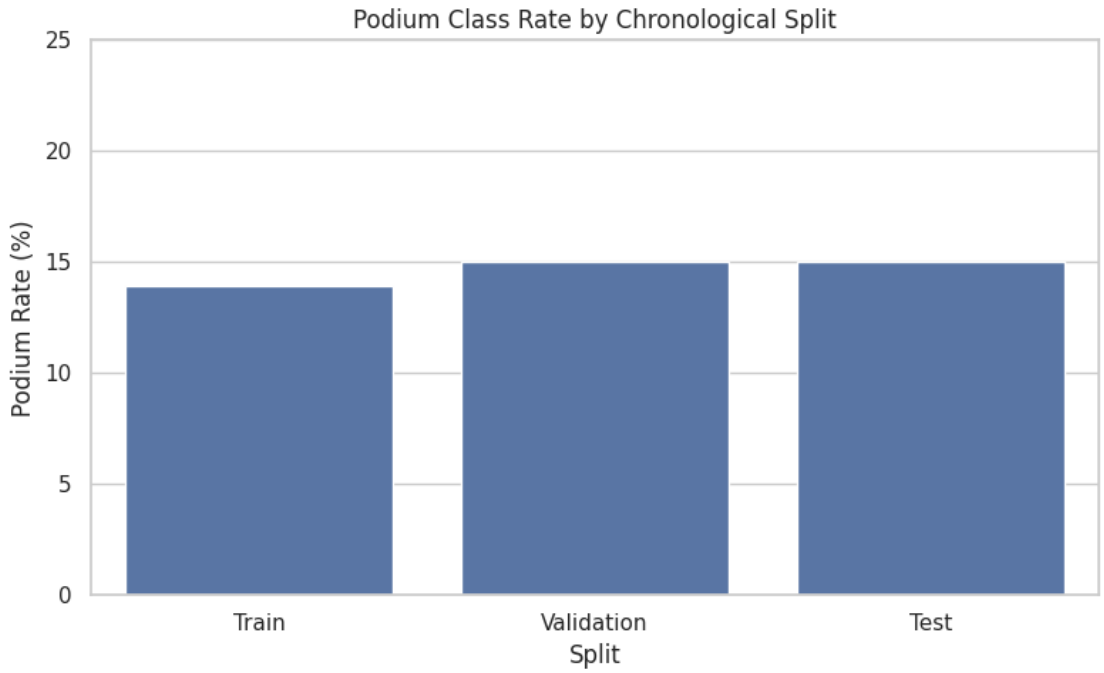


Figure A27. Positive-class prevalence by split. Whole-season chronological partitions have similar modern prevalence but different years and sample sizes. [3, cell 110] Original output: __results___109_0.png.

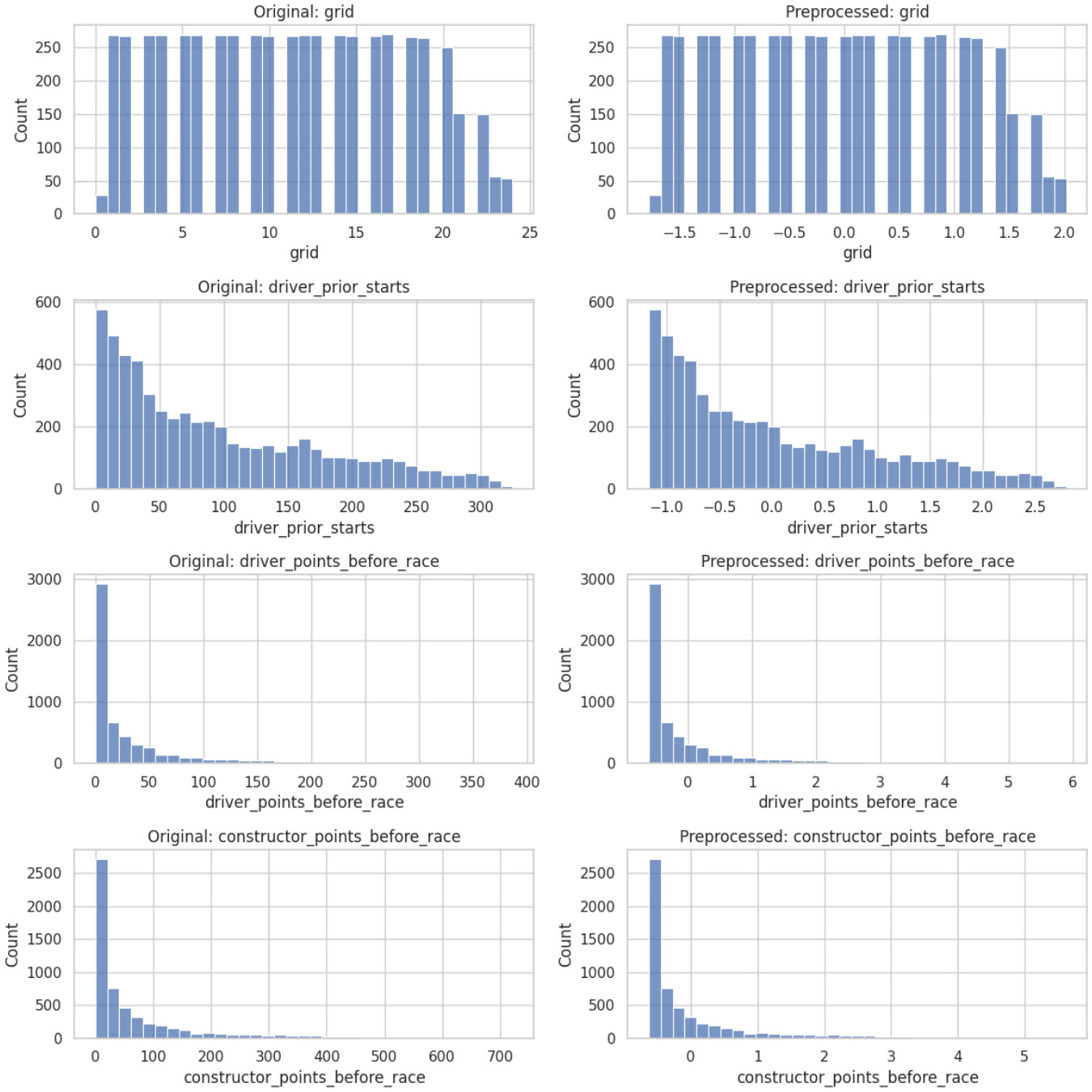


Figure A28. Training feature distributions before and after preprocessing. Median imputation and scaling change numerical center/range; they do not necessarily remove skew or establish comparability across sporting contexts. [3, cell 112] Original output: __results___111_0.png.

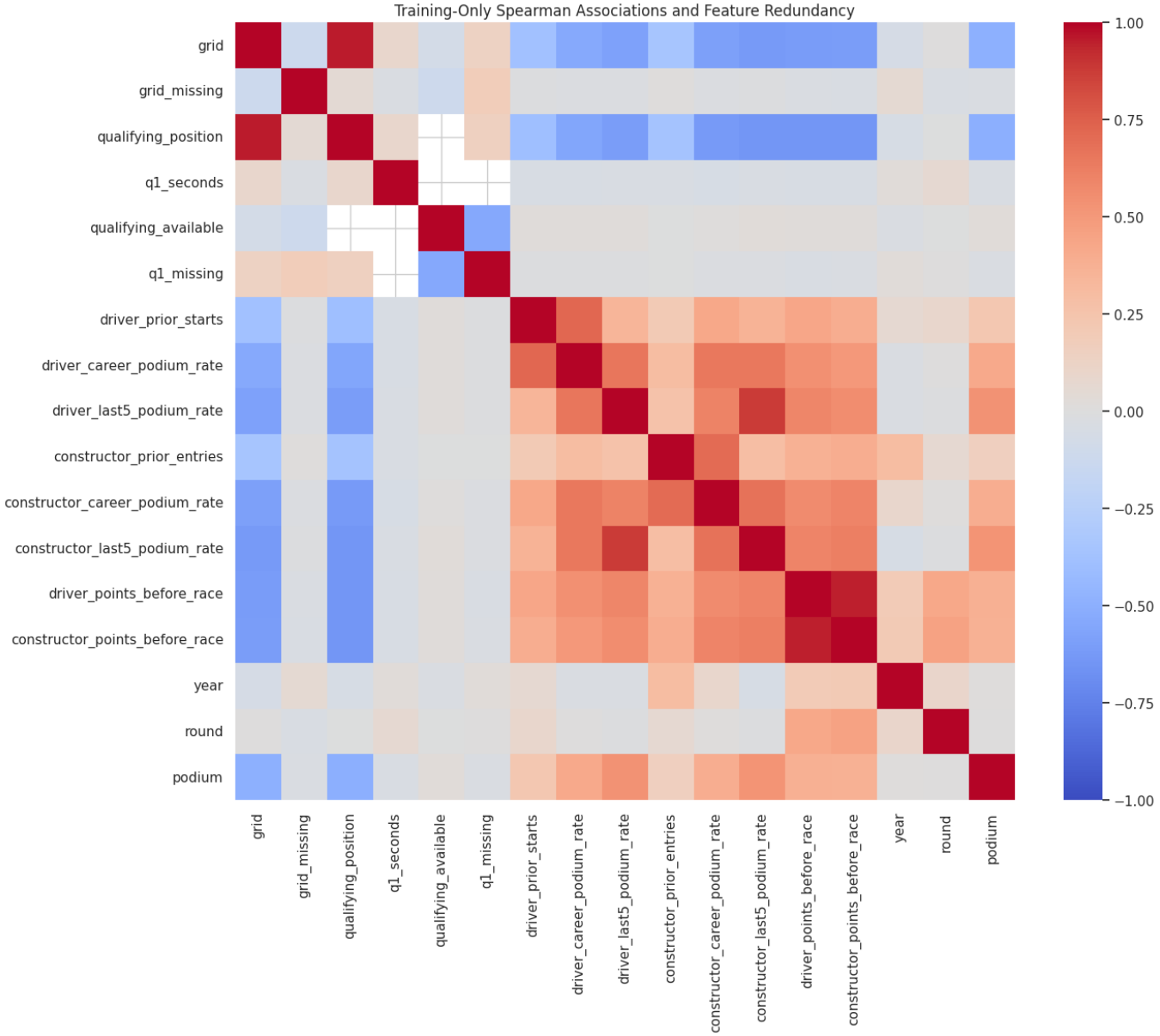


Figure A29. Training-only Spearman associations. Feature-target associations and feature redundancy provide training-only evidence; correlation is not causation and does not prove temporal validity. [3, cell 112] Original output: __results___111_1.png.

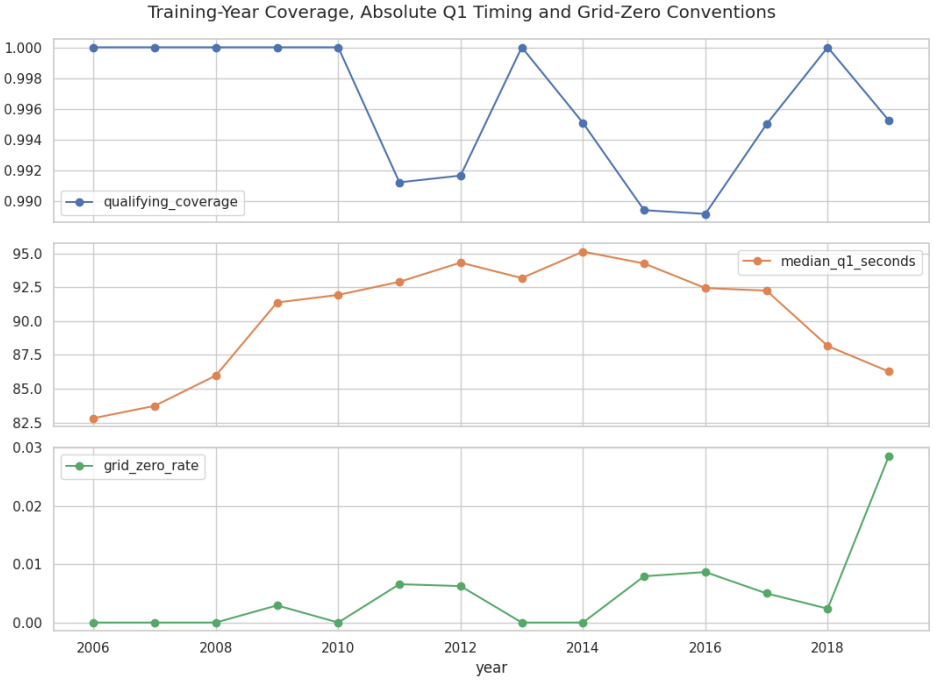


Figure A30. Training-year coverage, Q1 and grid-zero diagnostics. Temporal changes in availability, absolute timing and zero-grid conventions help qualify year and Q1 interpretations. [3, cell 112] Original output: __results___111_3.png.



Figure A31. Neural-network learning history. The actual run stops after 35 epochs and restores epoch 25; loss and PR-curve history are development diagnostics, with Keras PR-AUC distinct from table AP. [3, cell 121] Original output: __results___120_0.png.



Figure A32. Three-model validation metric comparison. Validation AP selects the NN; threshold-dependent scores use model-specific validation thresholds. [3, cell 124] Original output: __results___123_0.png.

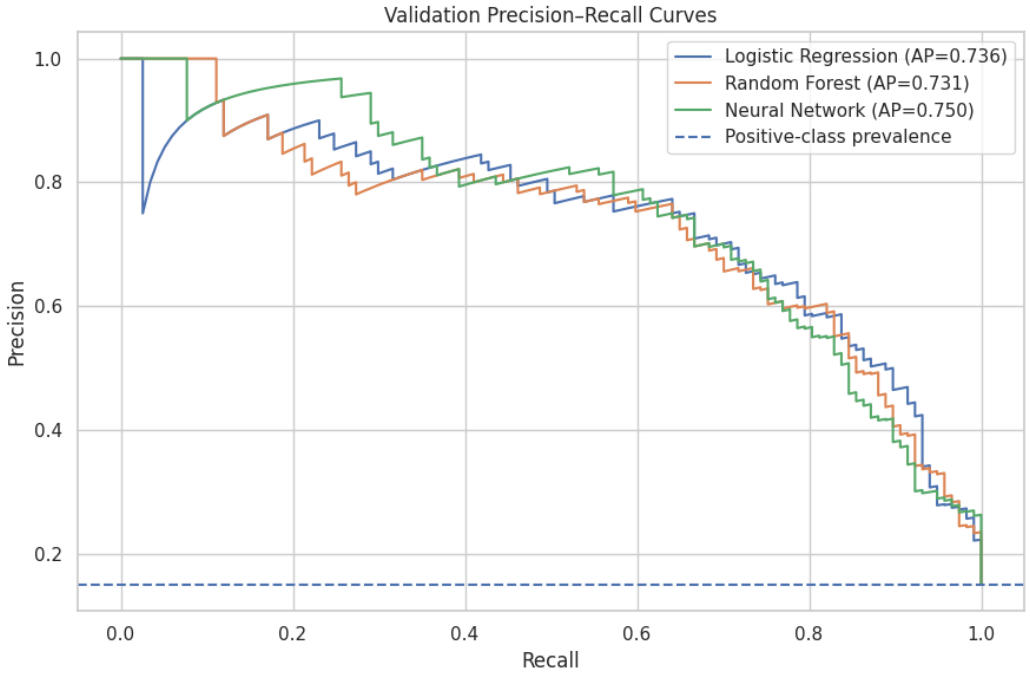


Figure A33. Validation precision-recall curves. The curves show precision/recall trade-offs; they complement a single AP or F1 value under class imbalance. [3, cell 125] Original output: __results___124_0.png.

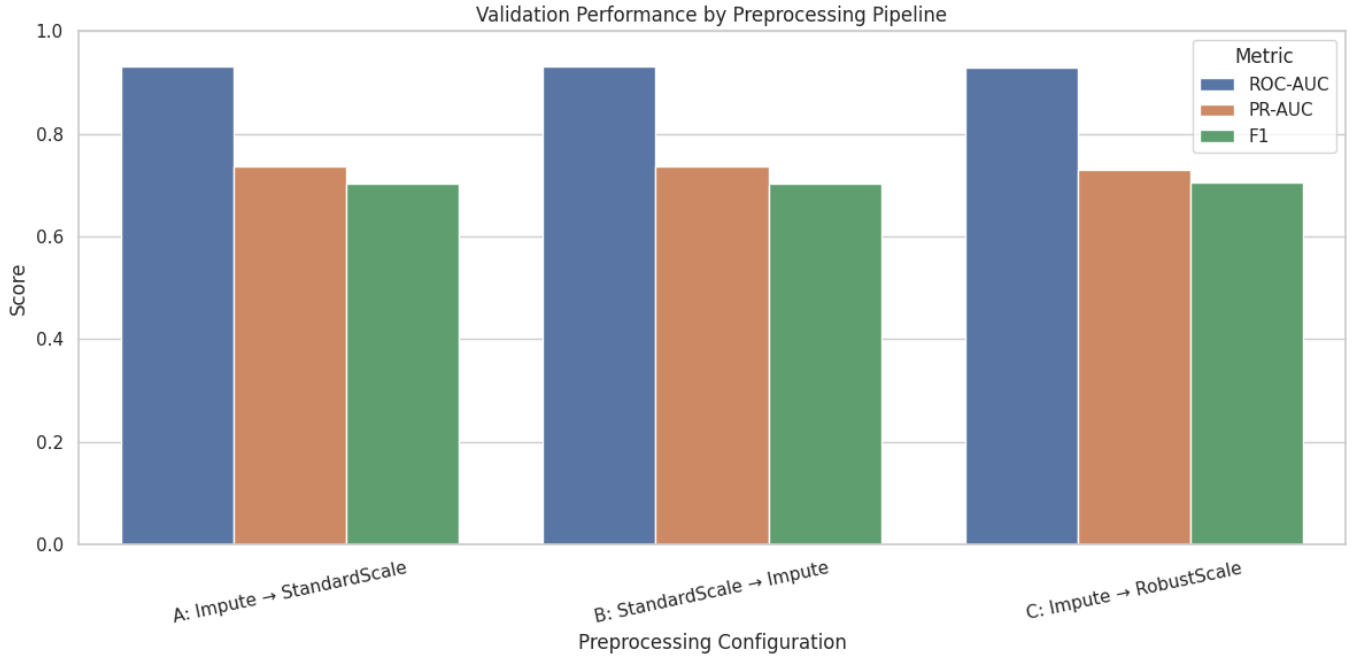


Figure A34. Original preprocessing comparison. Order A/B and robust-scaling reference C achieve close validation AP; they do not establish identical transformations. [3, cell 133] Original output: __results___132_0.png.

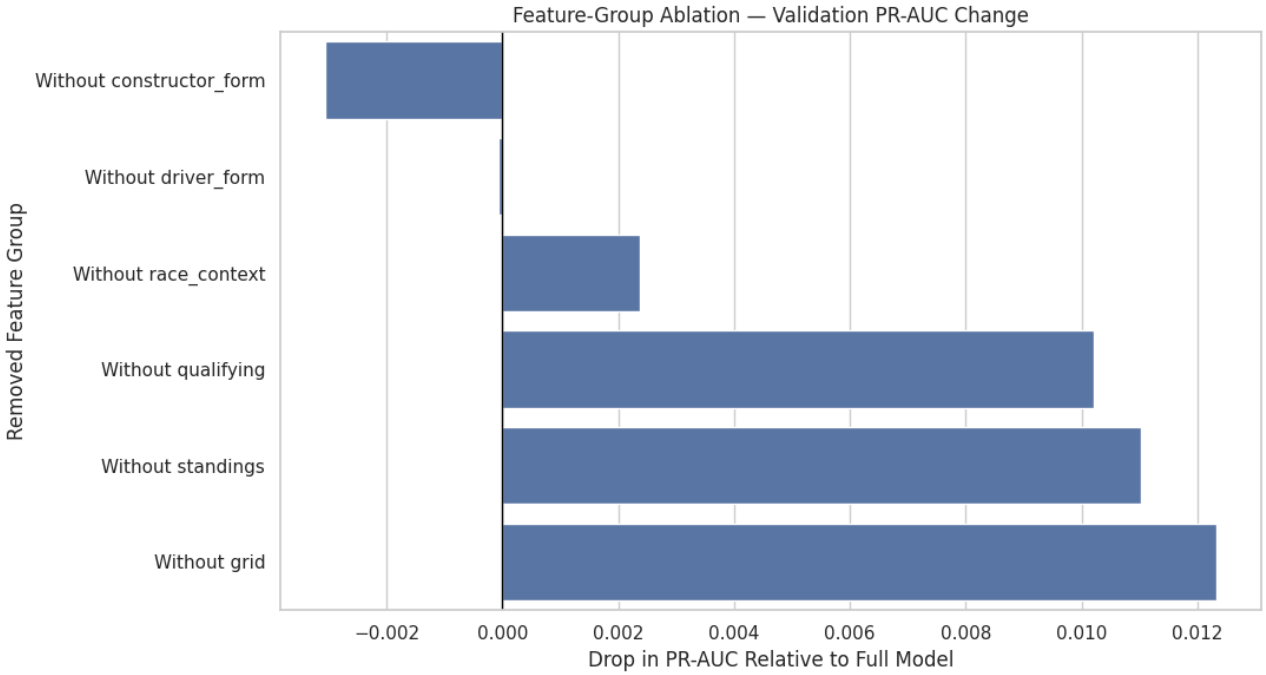


Figure A35. Feature-group ablation. Removing grid, standings or qualifying hurts LR validation AP most; correlated group information limits isolated-importance claims. [3, cell 135] Original output: __results___134_0.png.

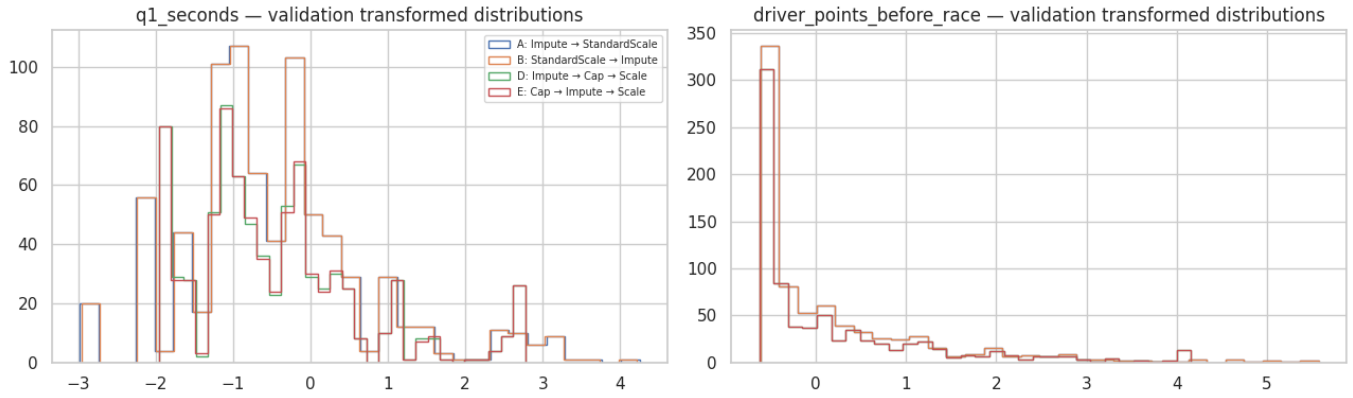


Figure A36. Transformed distributions for both order experiments. The displayed transformed Q1 and point distributions expose numerical differences even where rounded aggregate metrics agree. [3, cell 137] Original output: __results___136_5.png.

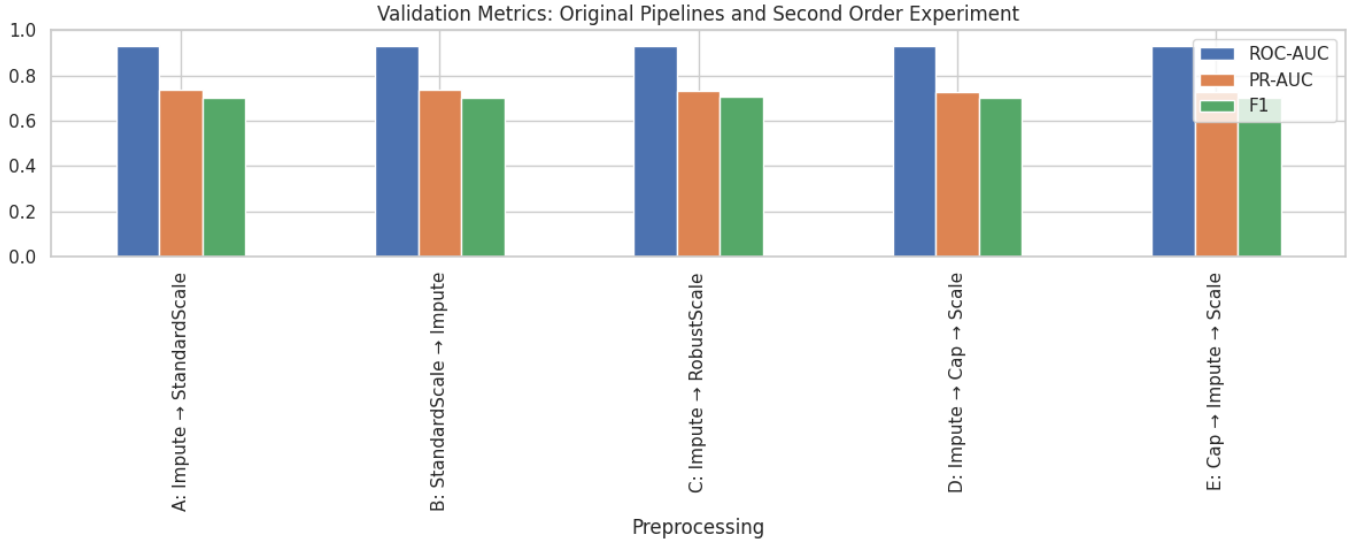


Figure A37. Validation metrics across original and second order pipelines. Capping/imputation orders do not improve validation AP over the original reference; no gain is assumed. [3, cell 137] Original output: __results___136_6.png.

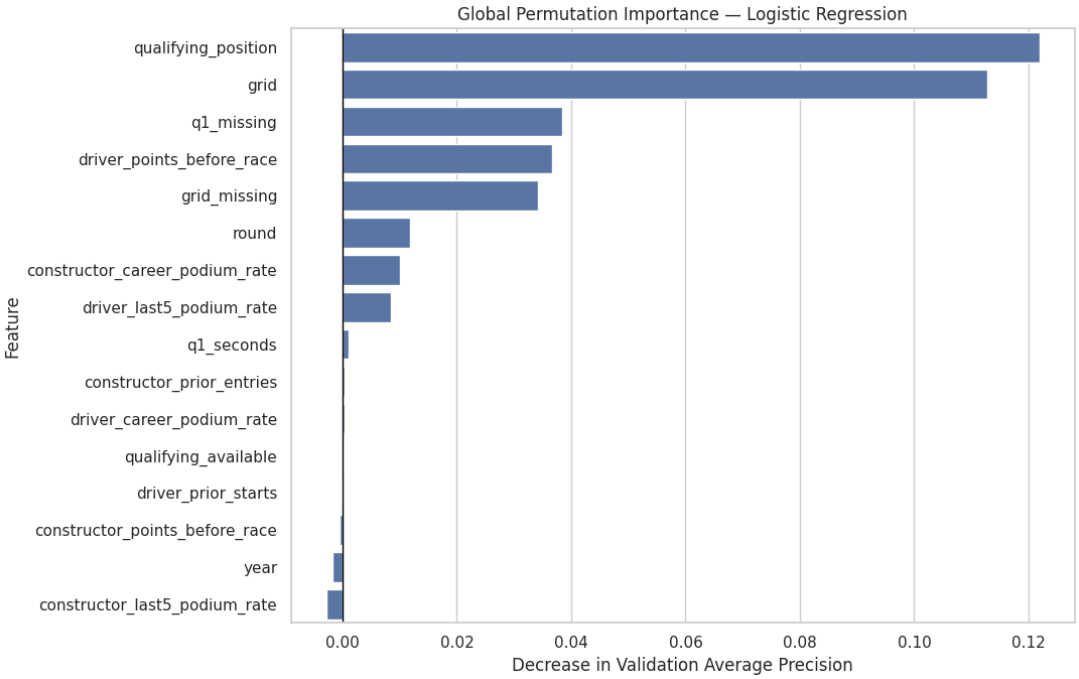


Figure A38. Global permutation importance, Logistic Regression. Qualifying position and grid produce the largest mean validation AP losses under independent shuffling; correlated features and repeat variability affect interpretation. [3, cell 140] Original output: __results___139_1.png.

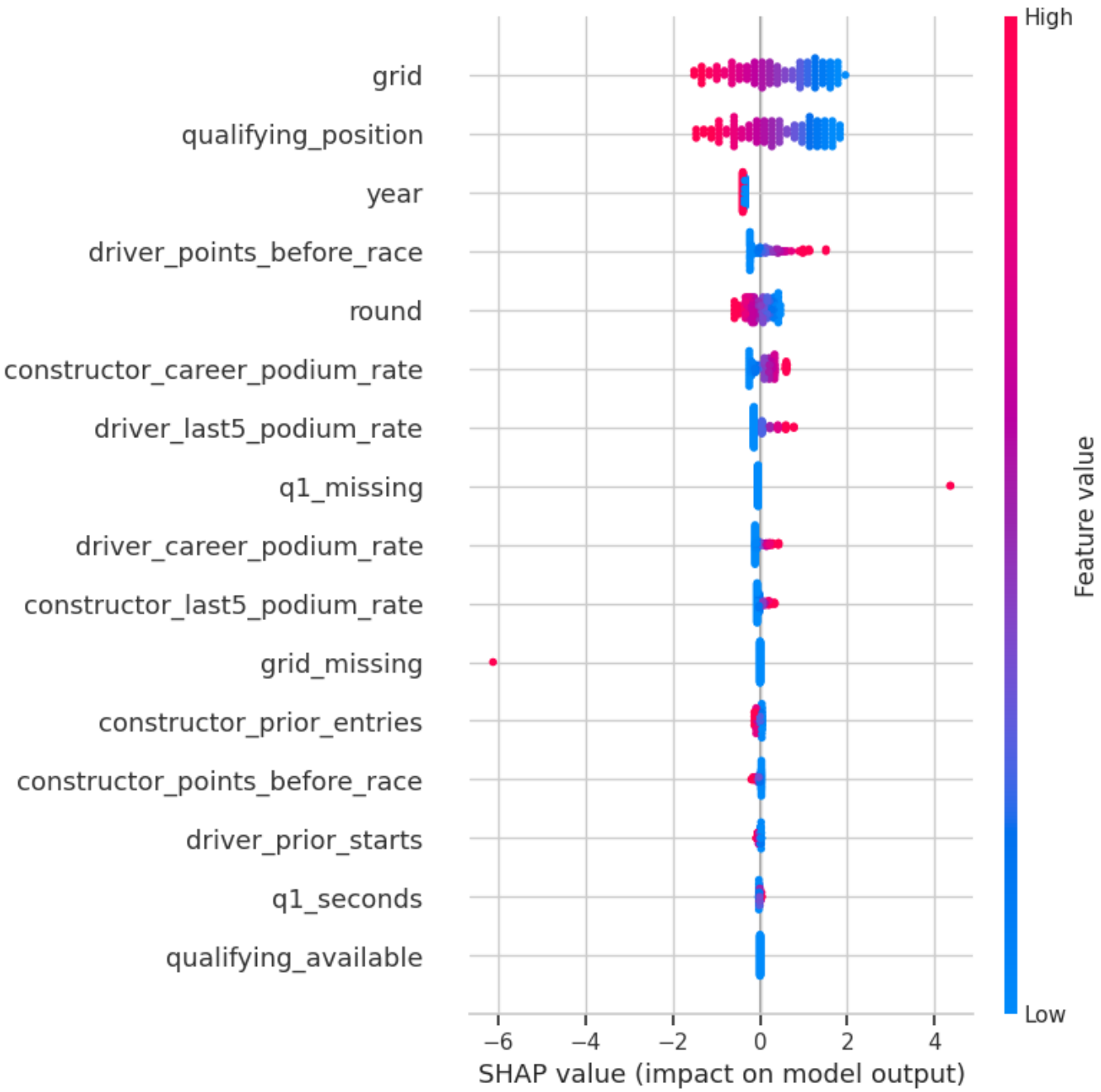


Figure A39. Global SHAP summary, Logistic Regression. SHAP explains log-odds relative to a training background. Feature patterns reflect fitted model reliance, not intervention effects. [3, cell 141] Original output: __results___140_2.png.

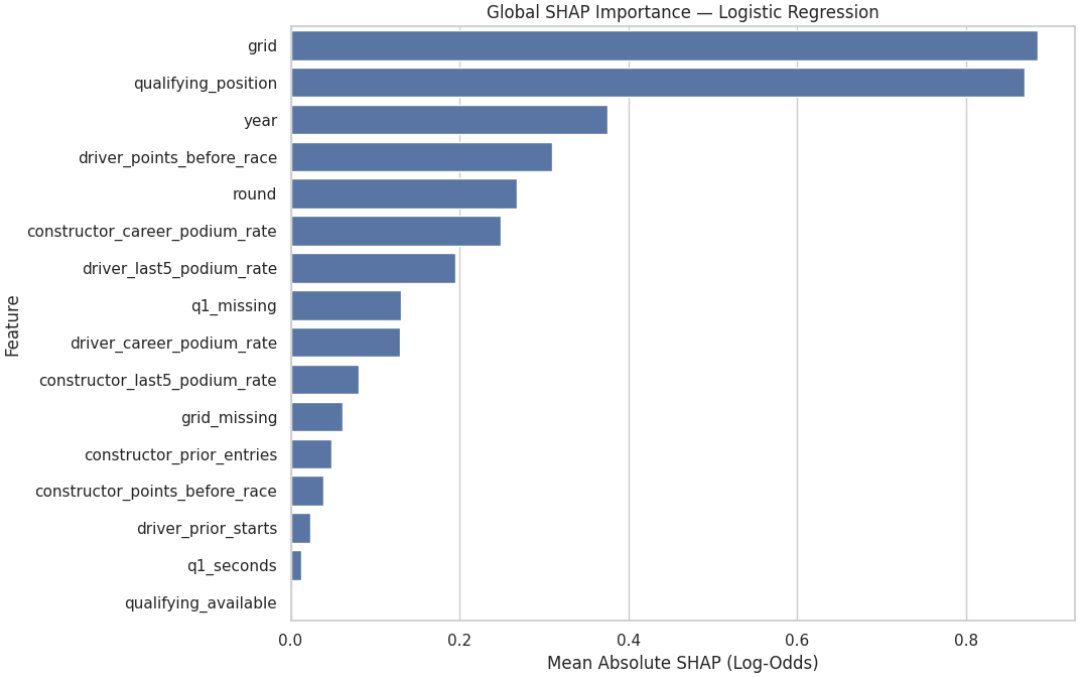


Figure A40. Mean absolute SHAP importance. Grid and qualifying position dominate the displayed mean absolute attribution; this is a different quantity from permutation AP sensitivity. [3, cell 142] Original output: __results___141_1.png.

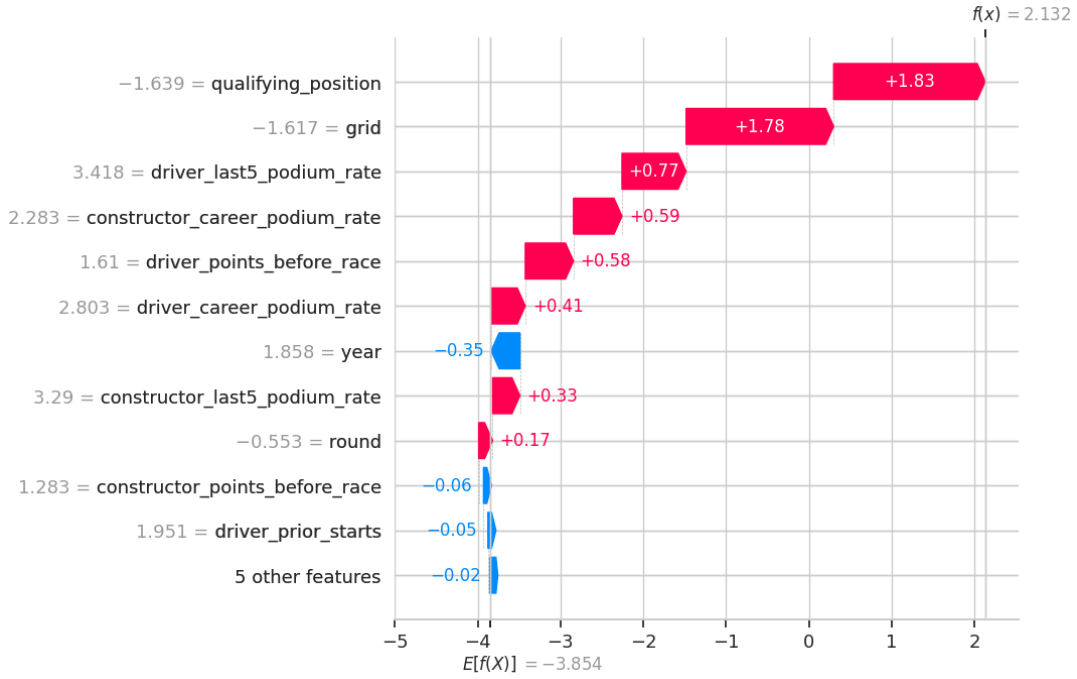


Figure A41. Local SHAP waterfall. The validation example (driverId 1, raceId 1037) shows additive log-odds contributions relative to the background; it does not explain the NN. [3, cell 143] Original output: __results___142_1.png.

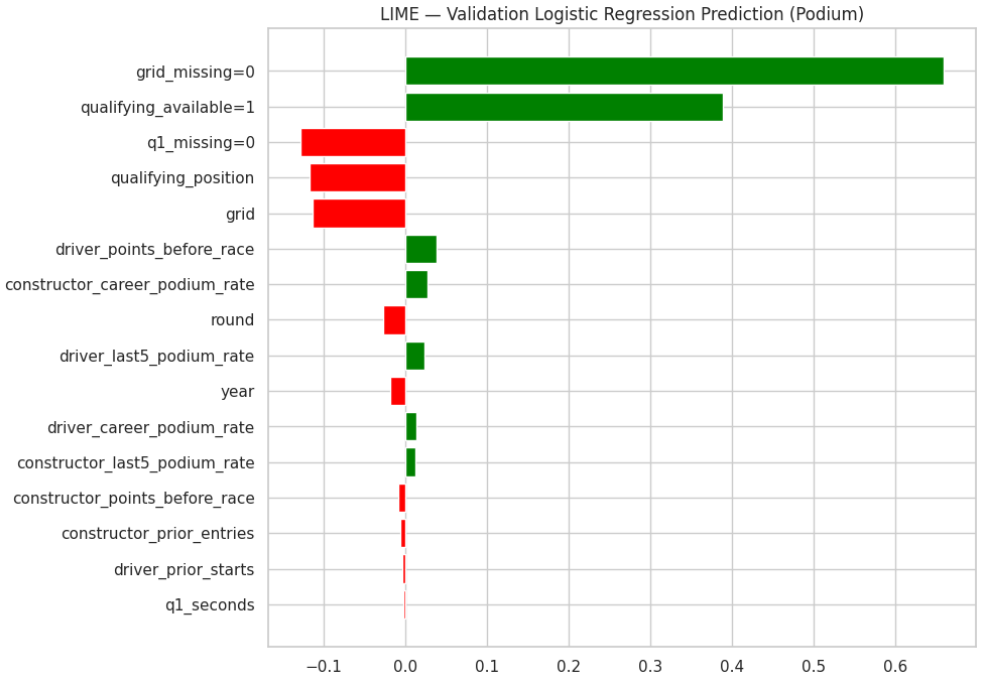


Figure A42. Local LIME explanation. The same validation record is explained with 1,500 perturbations. Local weighted surrogate R-squared is 0.8082; unrealistic correlated-feature perturbations limit interpretation. [3, cell 143] Original output: __results___142_2.png.

Appendix B. Requirement coverage and release checks

Requirement	Report evidence	Status
Prediction objective and cutoff	Sections 1 and 8	Covered
14 relational tables, PK/FK and immutable sources	Sections 2 and 4; final audit	Covered; all assertions passed
EDA before/after cleaning and preprocessing impact	Sections 3-4 and 9; Figures A1-A14, A26-A30	Covered
Three multi-hop engineering answers with figures	Sections 5-7; Figures A15-A25	Covered
Feature rationale, derivation and past-only histories	Section 8; Figures A26, A29-A30	Covered; historical limits disclosed
Chronological split and same-row model comparison	Sections 9-11	Covered
LR, RF, shallow NN; required metrics on all splits	Sections 10-11; Figures A31-A33	Covered; all 9 reports
Two preprocessing-order experiments	Section 12; Figures A34, A36-A37	Covered
Feature-group ablation	Section 13; Figure A35	Covered; six removals plus full reference
Permutation, global SHAP, local SHAP and LIME	Section 14; Figures A38-A42	Covered
Complete and partially missing inference	Section 15	Both executed
All notebook figures in report	Appendix A	42 included; 0 missing
Historical/model/methodological limitations	Sections 3-16	Disclosed
Evaluator visibility and assignment submission	Public setting observed; separate submission	Sharing unchanged; no submission

Numerical reconciliation. Tables and numerical claims were checked against the saved Version 13 execution evidence. The report introduces no replacement model results, invented confidence estimates or newly trained comparisons. Figures are the original saved output images. The original figure sequence and image SHA-256 values are recorded in the build evidence to detect omissions or substitutions.

Version 13 reconciliation. All 125 Python source cells and all numerical output tables match Version 12. The exact NN best validation loss is refreshed from this run; tiny floating-point differences do not alter the displayed comparison metrics. All 42 figures were retrieved from the saved Version 13 artifact, and report tables were rebuilt from Version 13 outputs. Kaggle reports a successful 127.6-second run. The released notebook opening Markdown was corrected to 127.6 seconds to match this verified run. No other numerical mismatch was identified at the displayed precision.

Unresolved issues. There is no missing report result or figure. The historical cause of the Q1 outlier, exact race-start eligibility for every historical entry, causal retirement causes, official as-of standings equivalence, probability calibration and the previously inspected test period remain substantive limitations. The public sharing setting was observed; independent signed-out access, any required GitHub packaging and actual assignment submission were not performed by this report-generation action.